ACS NANO

www.acsnano.org

ARTICLE

Brain Nucleic Acid Delivery and Genome Editing via Focused Ultrasound-Mediated Blood−Brain Barrier Opening and Long-Circulating Nanoparticles

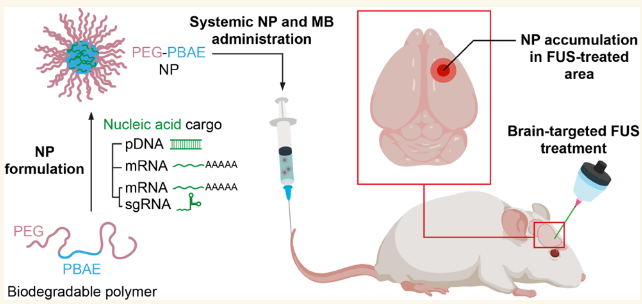
Gijung Kwak, Angad Grewal, Hasan Slika, Griffin Mess, Haolin Li, Mohit Kwatra, Alexandros Poulopoulos, Graeme F. Woodworth, Charles G. Eberhart, Han Seok Ko, Amir Manbachi, Justin Caplan, Richard J. Price, Betty Tyler, and Jung Soo Suk*

Cite This: *ACS Nano* 2024, 18, 24139−24153

Read Online

ACCESS | Metrics & More | Article Recommendations | Supporting Information

ABSTRACT: **We introduce a two-pronged strategy comprising focused ultrasound (FUS)-mediated blood−brain barrier (BBB) opening and long-circulating biodegradable nanoparticles (NPs) for systemic delivery of nucleic acids to the brain. Biodegradable poly(β-amino ester) polymer-based NPs were engineered to stably package various types of nucleic acid payloads and enable prolonged systemic circulation while retaining excellent serum stability. FUS was applied to a predetermined coordinate within the brain to transiently open the BBB, thereby allowing the systemically administered long-circulating NPs to traverse the BBB and accumulate in the FUS-treated brain region, where plasmid DNA or mRNA payloads produced reporter proteins in astrocytes and neurons. In contrast, poorly circulating and/or serum-unstable NPs, including the lipid NP analogous to a platform used in clinic, were unable to provide efficient nucleic acid delivery to the brain regardless of the BBB-opening FUS. The marriage of FUS-mediated BBB opening and the long-circulating NPs engineered to copackage mRNA encoding CRISPR-associated protein 9 and single-guide RNA resulted in genome editing in astrocytes and neurons precisely in the FUS-treated brain region. The combined delivery strategy provides a versatile means to achieve efficient and site-specific therapeutic nucleic acid delivery to and genome editing in the brain via a systemic route.**

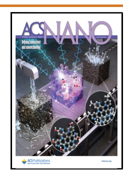


KEYWORDS: *long-circulating nanoparticle, blood−brain barrier, systemic nucleic acid delivery, brain gene therapy, biodegradable polymer*

INTRODUCTION

Nucleic acid therapy, such as gene therapy and editing, has emerged as a powerful therapeutic modality to potentially treat otherwise incurable brain disorders spanning malignant brain tumors to neurodegenerative diseases.[1−3] Successful nucleic acid therapy, regardless of the administration route, requires widespread therapeutic coverage throughout the lesioned areas within the brain.[4] We have previously demonstrated that nanoparticles (NPs) possessing dense surface-coatings with poly(ethylene glycol) (PEG) polymers minimize the adhesive interactions of NPs with the brain extracellular matrix (ECM) to enhance NP dispersion in the brain.[5−9] In particular, we showed that NPs comprising PEG-coated surface and biodegradable poly(β-amino ester) (PBAE) polymer/plasmid DNA (pDNA)-complexed core (PEG−PBAE/pDNA NPs) provided widespread transgene expression in healthy or tumor-bearing rodent brains following intracranial administration.[6,10] This localized administration method, however, involves an invasive surgical procedure that often causes deleterious effects, including infection, edema, and neuronal damage.[11] The limitation can be circumvented by systemic administration, but NPs must address a series of biological barriers, including avoidance of liver accumulation, efficient circulation to cerebral capillaries, and penetration through the blood−brain barrier

Received: April 21, 2024
Revised: August 12, 2024
Accepted: August 14, 2024
Published: August 22, 2024



https://doi.org/10.1021/acsnano.4c05270
ACS Nano 2024, 18, 24139−24153

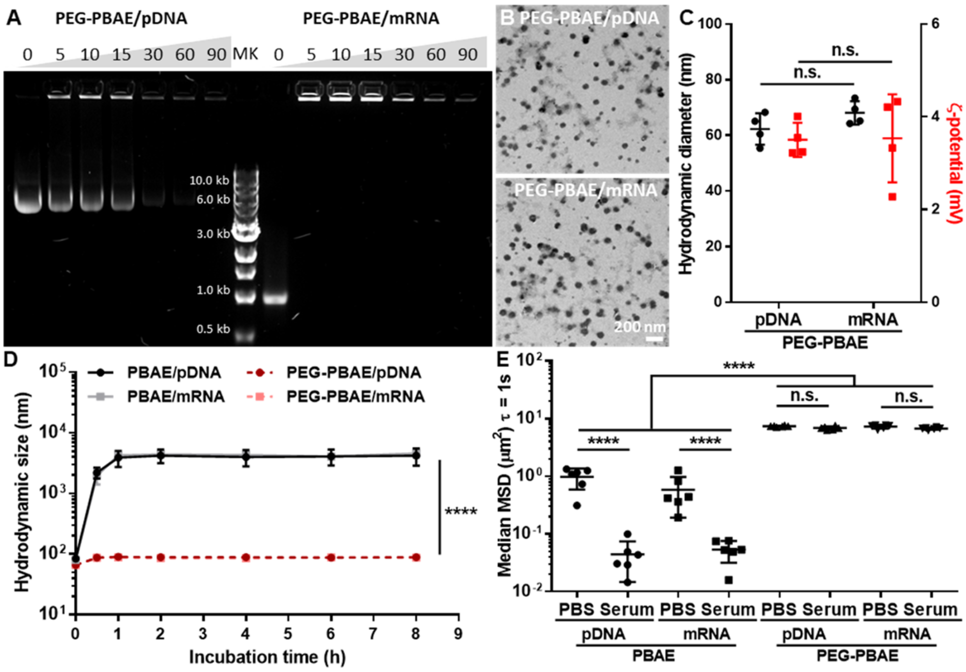


Figure 1. PEG−PBAE NPs, but not un-PEGylated PBAE NPs, carrying pDNA or mRNA retain colloidal stability in physiological conditions relevant to the systemic delivery of NPs to the brain. (A) Representative agarose gel electrophoresis migration assay showing robust packaging of pDNA (left) or mRNA (right) in PEG−PBAE NPs. Numbers indicate the polymer-to-nucleic-acid weight ratios. MK: molecular weight marker. (B) Representative transmission electron micrographs of PEG−PBAE NPs carrying pDNA (top) or mRNA (bottom). Scale bar = 200 nm. (C) Hydrodynamic diameters and ζ-potentials of PEG−PBAE NPs carrying pDNA or mRNA. (D) Colloidal stability of PBAE and PEG−PBAE NPs carrying pDNA or mRNA incubated in aCSF at 37 °C for up to 8 h. (E) Median MSD values of PBAE and PEG−PBAE NPs carrying pDNA or mRNA in PBS or whole mouse serum. MSD is a square of distance traveled by an individual particulate matter within a predetermined time interval (i.e., time scale; τ = 1 s) and thus is directly proportional to the particle diffusion rate. n.s.: no significance, ****$p < 0.0001$ (one-way or two-way analysis of variance (ANOVA)).

(BBB), to reach target disease areas and cells within the brain.[12,13]

The BBB, a tightly sealed multicellular layer lining the cerebral capillaries,[12,13] is arguably the most challenging barrier to the delivery of therapeutics and NPs to the brain via the systemic route.[14] Molecular targeting and biochemical/osmotic modulation have been explored over the past few decades to enhance the delivery of NPs across the BBB, but the approaches entail several limitations, including suboptimal delivery efficacy, lack of targetability, and safety issues.[15,16] More recently, we and others have investigated the use of transcranial focused ultrasound (FUS) as a noninvasive means to open the BBB to promote the penetration of systemically administered NPs into the brain.[17−19] The technique involves perturbation of the BBB by activating intravascular gas-filled microbubbles (MBs; FDA-approved contrasting agents for ultrasound imaging) with FUS applied at a predetermined coordinate within the brain.[20] The MBs passing through the cerebral capillaries then oscillate stably in the FUS focal region and generate acoustic radiation force that acts on the vessel wall to open the BBB in a transient, reversible, and targeted manner.[18,21,22] Encouragingly, FUS-mediated BBB opening has been well-tolerated among patients with brain disorders in multiple clinical studies[23−26] and has been recently shown to significantly enhance the systemic delivery of albumin-bound paclitaxel (i.e., Abraxane) across the BBB in patients with recurrent glioblastoma.[27]

To fully exploit the FUS-mediated BBB opening and NP penetration, it is conceivable that systemically administered NPs should be able to circulate stably during the time window of the BBB opening. NPs will also need to retain the particle diameters small enough to fit in the vascular openings created by FUS to traverse the BBB. We hypothesized that PEG−PBAE NPs, due to the dense surface PEG coating, would resist nonspecific adhesive interactions with hepatic cells and serum proteins to stably circulate and retain colloidal stability following systemic administration. We thus engineered PEG−PBAE NPs to carry either pDNA or mRNA, extensively characterized the physicochemical properties and physiological stability in vitro, and investigated their ability to stably circulate after system administration in vivo. We then tested whether FUS-mediated BBB opening promoted accumulation of systemically administered PEG−PBAE NPs and production of reporter proteins encoded by respective nucleic acid payloads precisely in the FUS-treated brain region. Finally, we engineered PEG−PBAE NPs to copackage mRNA-encoding clustered regularly interspaced palindromic repeats (CRISPR)-associated protein 9 (Cas9) and single-guide RNA (sgRNA) and investigated the ability of the NPs, in conjunction with FUS, to mediate somatic genome editing in the brain in vivo.

RESULTS

PEG−PBAE NPs Retain the Physicochemical Properties and Colloidal Stability in Physiological Conditions Relevant to Systemic NP Delivery to the Brain. PBAE and PEG−PBAE polymers were synthesized and characterized as we previously reported.[6] To engineer PEG−PBAE NPs carrying various nucleic acid payloads, we compacted either pDNA or mRNA with a blend of PBAE and PEG−PBAE

Table 1. Physicochemical Properties of NPs

NP	hydrodynamic diameter[a] ± SD (nm)	PDI[a] ± SD	ζ-potential[b] ± SD (mV)
PBAE/pDNA	73.8 ± 3.1	0.23 ± 0.02	18.4 ± 3.2
PBAE/mRNA	70.4 ± 2.6	0.25 ± 0.03	19.8 ± 2.2
PEG−PBAE/pDNA	61.4 ± 2.5	0.21 ± 0.03	2.1 ± 0.3
PEG−PBAE/mRNA	64.8 ± 3.1	0.19 ± 0.04	2.5 ± 0.4
PEG−PBAE/mRNA+sgRNA	62.2 ± 3.8	0.23 ± 0.03	2.2 ± 0.3
LNP/mRNA	71.1 ± 3.5	0.39 ± 0.03	−3.1 ± 1.7

[a]Hydrodynamic diameter (number mean) and polydispersity index (PDI) were measured by dynamic light scattering (DLS) in 10 mM NaCl at pH 7.4. Mean ± SD (N = 4). [b]ζ-Potential was measured by laser Doppler anemometry in 10 mM NaCl at pH 7.0. Mean ± SD (N = 4)

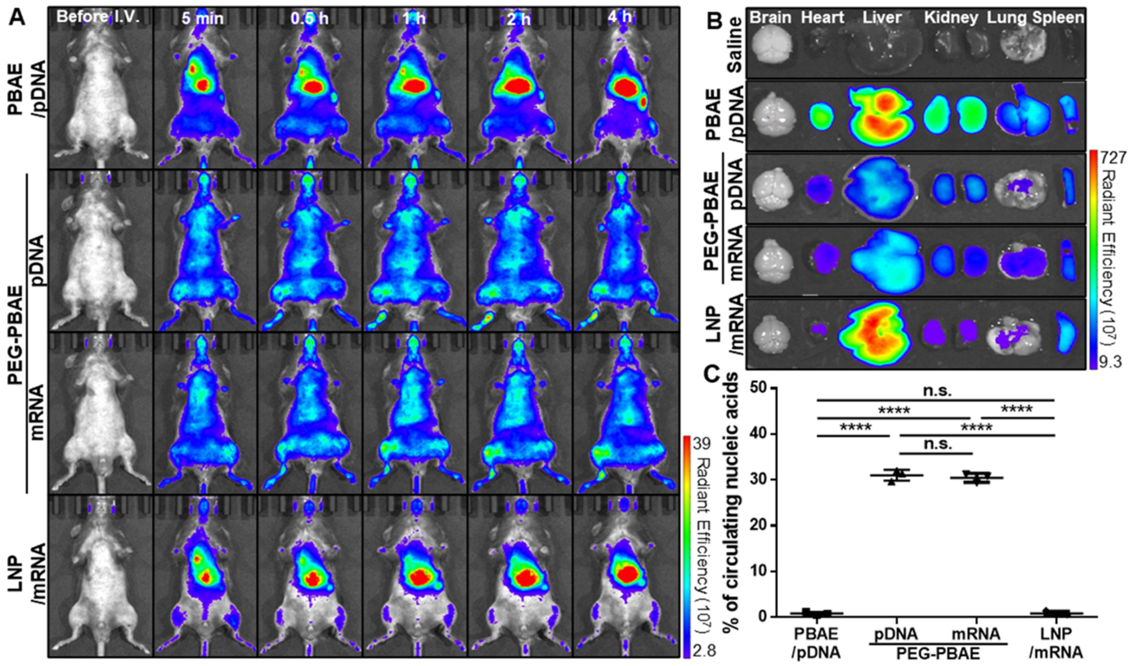


Figure 2. PEG−PBAE NPs, but not clinically used LNPs, carrying pDNA or mRNA, exhibit long circulation in the bloodstream following systemic administration. (A) Representative whole-body fluorescence images of live animals intravenously treated with PBAE NPs, PEG−PBAE NPs, or LNPs carrying Cy5-labeled pDNA or mRNA over time (N = 3 animals per group). (B) Representative fluorescence images of major organs harvested at 4 h postadministration of NPs from animals as in panel (A). (C) Percentage of circulating nucleic acids determined by quantifying the Cy5 fluorescence intensity of the serum collected at 4 h postadministration of NPs from animals intravenously treated with PBAE NPs, PEG−PBAE NPs, or LNPs carrying Cy5-labeled pDNA or mRNA. n.s.: no significance, ****p < 0.0001 (one-way ANOVA).

polymers over a range of polymer-to-nucleic-acid weight ratios. We confirmed with a gel electrophoresis migration assay that pDNA or mRNA was fully compacted by the polymer mixture at the polymer-to-nucleic-acid weight ratio of 60:1 to form pDNA- or mRNA-loaded PEG−PBAE NPs (PEG−PBAE/pDNA or PEG−PBAE/mRNA NPs, respectively) (Figure 1A). In parallel, un-PEGylated PBAE NPs carrying pDNA or mRNA (PBAE/pDNA or PBAE/mRNA NPs, respectively) were prepared with PBAE polymers only at the same weight ratio. Transmission electron microscopy (TEM) revealed that PEG−PBAE NPs, regardless of the type of nucleic acid payloads, exhibited spherical morphology and <100 nm geometric diameters (Figure 1B). PEG−PBAE/pDNA and PEG−PBAE/mRNA NPs showed comparable hydrodynamic diameters of 61.4 ± 2.5 and 64.8 ± 3.1 nm and ζ-potentials of 2.1 ± 0.3 and 2.5 ± 0.4 mV, respectively (Figure 1C and Table 1). In comparison, PBAE/pDNA and PBAE/mRNA NPs were slightly larger than their respective PEGylated NP counterparts with average hydrodynamic diameters of ∼70 nm and possessed positively charged surfaces with ζ-potentials of ∼20 mV (Table 1). We next assessed particle colloidal stability in artificial cerebrospinal fluid (aCSF) by monitoring the particle hydrodynamic diameters over time at 37 °C where we found that PEG−PBAE NPs, regardless of the payload type, retained the hydrodynamic diameter <100 nm at least up to 8 h (Figure 1D). In contrast, PBAE/pDNA and PBAE/mRNA NPs instantaneously aggregated in aCSF, and the particle hydrodynamic diameters reached several microns as early as 30 min after the incubation (Figure 1D). To investigate the serum colloidal stability of these NPs, we conducted multiple particle tracking (MPT) analysis. MPT measures the mean square displacement (MSD) values of NPs in biological medium, such as phosphate buffer saline (PBS) and whole mouse serum. MSD is a square of distance traveled by an individual NP and is directly proportional to the particle diffusion rate, which is inversely proportional to the particle diameter based on the Stokes−Einstein relationship.[28] We found that the MSD values of PEG−PBAE NPs were virtually identical in PBS and mouse serum after a 30 min incubation regardless of the payload type, indicating that particle hydrodynamic diameters were un-

https://doi.org/10.1021/acsnano.4c05270
ACS Nano 2024, 18, 24139−24153

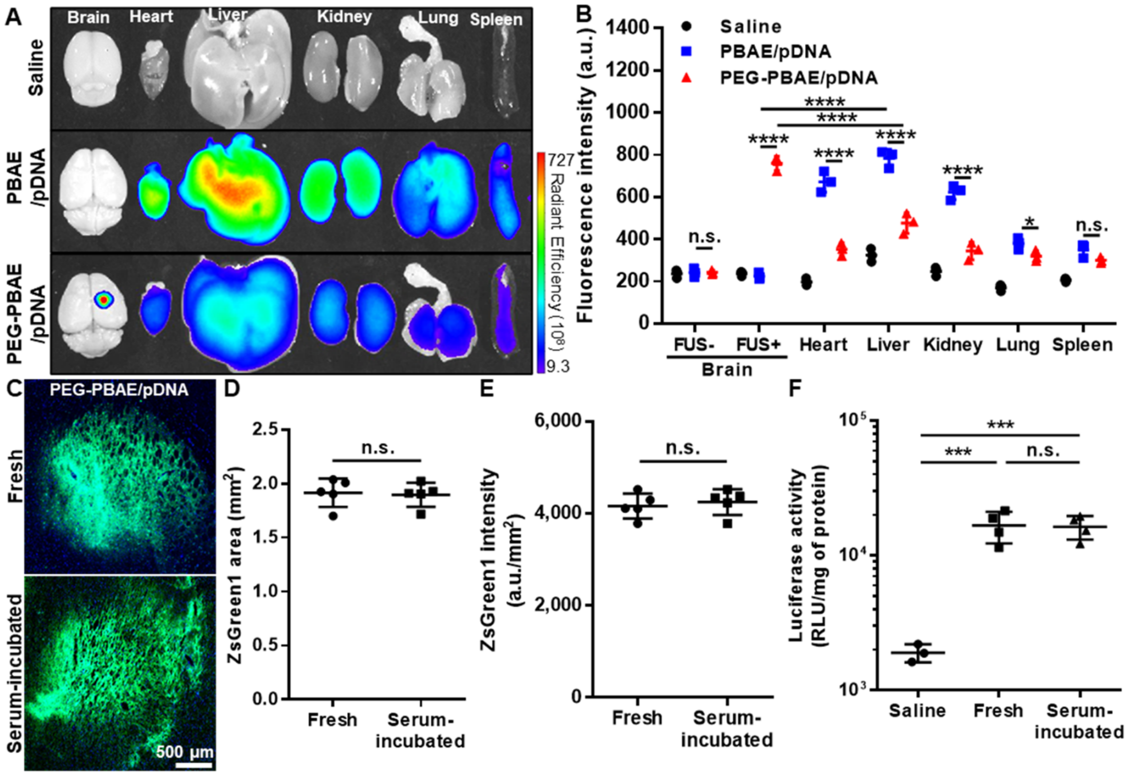


Figure 3. PEG−PBAE NPs carrying pDNA accumulate in the FUS-treated brain region far greater than other major organs following systemic administration and retain the ability to mediate widespread transgene expression in the brain following serum incubation and subsequent intracranial administration. (A) Representative fluorescence images and (B) Cy5 fluorescence intensity of brains and other major organs harvested at 4 h postadministration of NPs from animals intravenously treated with saline or PBAE or PEG−PBAE NPs carrying Cy5-labeled pDNA and subsequently received a FUS treatment on the right striatum (*N* = 3 animals per group). (C) Representative fluorescence images showing reporter ZsGreen1 expression in the brain sections at the infusion site within healthy mouse brain striatum 48 h after the intracranial administration of freshly prepared or serum-incubated PEG−PBAE NPs carrying ZsGreen1-expressing pDNA at a pDNA dose of 0.1 mg/kg (*N* = 5 animals per group). Blue: nucleus; green: ZsGreen1. Image-based quantification for (D) area and (E) intensity of ZsGreen1 expression. (F) Luciferase activity of homogenized brain tissues treated with freshly prepared or serum-incubated PEG−PBAE NPs carrying luciferase-expressing pDNA at a pDNA dose of 0.1 mg/kg via intracranial administration. n.s.: no significance, ****p* < 0.05, ****p* < 0.001, and *****p* < 0.0001 (one-way ANOVA).

changed in mouse serum (Figure 1E). In contrast, the MSD values of both PBAE NP formulations were over an order of magnitude lower in mouse serum compared to those in PBS, which reflects particle aggregation in mouse serum (Figure 1E).

PEG−PBAE NPs Exhibit Prolonged Circulation Following Systemic Administration. We next investigated the ability of PEG−PBAE NPs to stably circulate in the bloodstream following systemic administration. We treated male C57BL/6 mice with PBAE NPs, PEG−PBAE NPs, or lipid NPs (LNPs) carrying Cy5-labeled pDNA or mRNA at a nucleic acid dose of 0.5 mg/kg via tail vein injection and conducted live-animal whole-body fluorescence imaging for up to 4 h. Of note, the LNP formulation tested in this study is analogous to an intramuscular mRNA-based COVID-19 vaccine used in the clinic (i.e., Comirnaty). PEG−PBAE/pDNA and PEG−PBAE/mRNA NPs exhibited relatively uniform whole-body fluorescence distribution without noticeable attenuation of the intensity over time (Figure 2A). In contrast, fluorescence signals were rapidly accumulated in the livers of animals treated with PBAE/pDNA NPs or LNPs as early as 5 min after the administration, which persisted and were intensified over time (Figure 2A). Major organs and blood samples were harvested from these animals at 4 h postadministration of NPs to determine the residual amounts of NPs in each compartment. Consistent with the outcomes of the live-animal imaging study (Figure 2A), strong fluorescence was observed in the livers of animals treated with PBAE NPs or LNPs, but the intensity was markedly lower in those of animals treated with PEG−PBAE NPs (Figure 2B). On the other hand, while the fluorescence intensity measured in the serum samples harvested from PBAE NP-, LNP-, and saline-treated animals were virtually identical, the levels were markedly and equally greater for animals treated with PEG−PBAE NPs regardless of the payload type (Figure 2C). Quantitatively, 0.8 ± 0.3, 31.0 ± 1.2, 30.4 ± 1.0, and 0.9 ± 0.5%, respectively, of the systemically administered PBAE/pDNA NPs, PEG−PBAE/pDNA NPs, PEG−PBAE/mRNA NPs, and LNPs remained circulating in the bloodstream at 4 h postadministration of NPs (Figure 2C).

FUS-Mediated BBB Opening Promotes Accumulation of PEG−PBAE NPs in the Brain Following Systemic Administration. We hypothesized that the safe BBB opening by FUS would result in vascular openings with a certain size cutoff. To test this, we engineered densely PEGylated NPs with defined particle diameters. Commercially available red-fluorescent, carboxylated polystyrene (PS) NPs (i.e., Fluosphere) possessing different particle diameters, 40, 100, or 200 nm, were densely coated with PEG polymers via chemical conjugation, and effective surface shielding was confirmed by

https://doi.org/10.1021/acsnano.4c05270
ACS Nano 2024, 18, 24139−24153

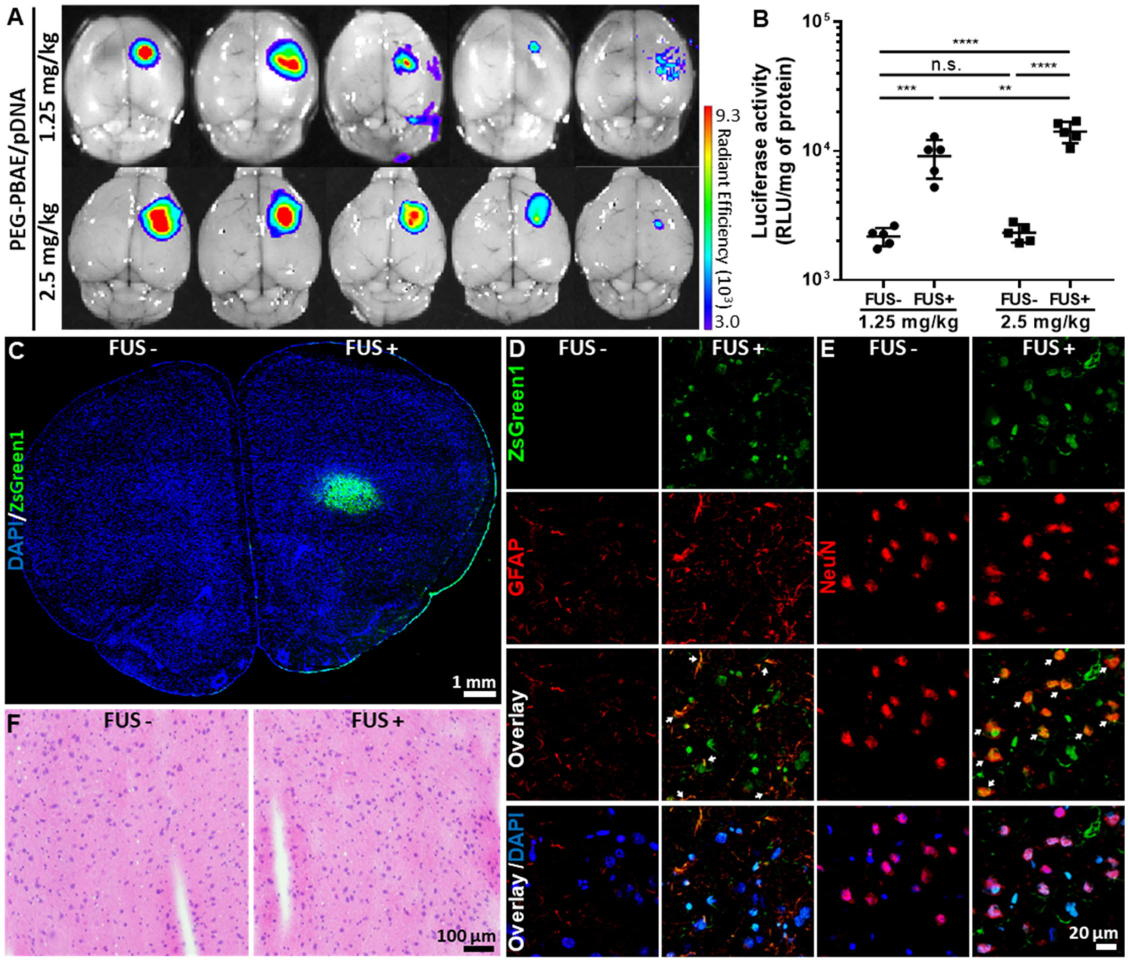


Figure 4. PEG−PBAE NPs carrying pDNA mediate reporter protein production in astrocytes and neurons precisely in the FUS-treated brain region following systemic administration. (A) Bioluminescence images of brains harvested at 48 h postadministration of NPs from animals intravenously treated with PEG−PBAE NPs carrying luciferase-expressing pDNA at a pDNA dose of 1.25 or 2.5 mg/kg and subsequently received a FUS treatment on right striata (N = 5 animals per group). (B) Quantification of luciferase activity in the homogenized hemispheres from the brains in panel (A). (C) Representative confocal micrograph of brain harvested at 48 h postadministration of NPs from animals intravenously treated with PEG−PBAE NPs carrying ZsGreen1-expressing pDNA at a pDNA dose of 2.5 mg/kg pDNA and subsequently received a FUS treatment on right striata (N = 5 animals per group). Blue: nucleus; green: ZsGreen1. Representative confocal micrographs showing reporter ZsGreen1 production in untreated or FUS-treated areas of (D) GFAP- and (E) NeuN-stained brains from animals identically treated as in panel (C). Blue: nucleus; green: ZsGreen1; red: astrocyte (GFAP) or neuron (NeuN). (F) Representative H&E-stained histological images of untreated or FUS-treated areas of the brain harvested from animals identically treated as in panel (C). n.s.: no significance, **p < 0.01, ***p < 0.001, and ****p < 0.0001 (one-way ANOVA).

near-neutral surface charges (i.e., ζ-potential) of NPs (Table S1). We then treated male C57BL/6 mice with a blend of single-sized PEGylated PS (PEG−PS) NPs and clinically used MBs (i.e., SonoVue) at doses of 5 mg/kg and 0.3 mL/kg, respectively, and immediately applied FUS to the right striatum. Of note, a single-point sonication was given using a 515-kHz transducer at a 0.6 MPa peak negative pressure with a 10 ms burst length for 2 min and a pulse repetition frequency of 1 Hz. Four hours after the administration, brain tissues were harvested from the treated animals and striatal NP accumulation was determined by whole-brain fluorescence imaging. We found that 40 and 100 nm PEG−PS NPs were accumulated precisely in the FUS-treated regions of the right striata with comparable fluorescence intensity, but negligible brain accumulation was observed with 200 nm of PEG−PS NPs (Figure S1).

We next repeated this study with PBAE and PEG−PBAE NPs carrying Cy5-labeled pDNA at a pDNA dose of 0.5 mg/kg. Four hours after the administration, we found that PEG−PBAE NPs were accumulated in the FUS-treated region of right striatum, similar to our observation with PEG−PS NPs as large as 100 nm in diameters (Figure S1), but PBAE NPs failed to do so (Figure 3A). We next measured the fluorescence intensity of tissue homogenates prepared with left and right brain hemispheres and other major organs harvested from the treated animals. PEG−PBAE NP-treated, but not PBAE NP-treated, animals exhibited strong fluorescence intensity in the FUS-treated (i.e., right) hemisphere (Figure 3B). Fluorescence intensity was markedly and significantly lower in major organs, including heart, liver, and kidney, harvested from PEG−PBAE NP-treated animals compared to PBAE NP-treated animals (Figure 3B), in agreement with our observation with whole-tissue imaging (Figure 3A). Notably, PEG−PBAE NP-treated animals exhibited approximately 5.5 times greater right brain hemisphere-to-liver ratio of fluorescence intensity compared with PBAE NP-treated animals.

https://doi.org/10.1021/acsnano.4c05270
ACS Nano 2024, 18, 24139−24153

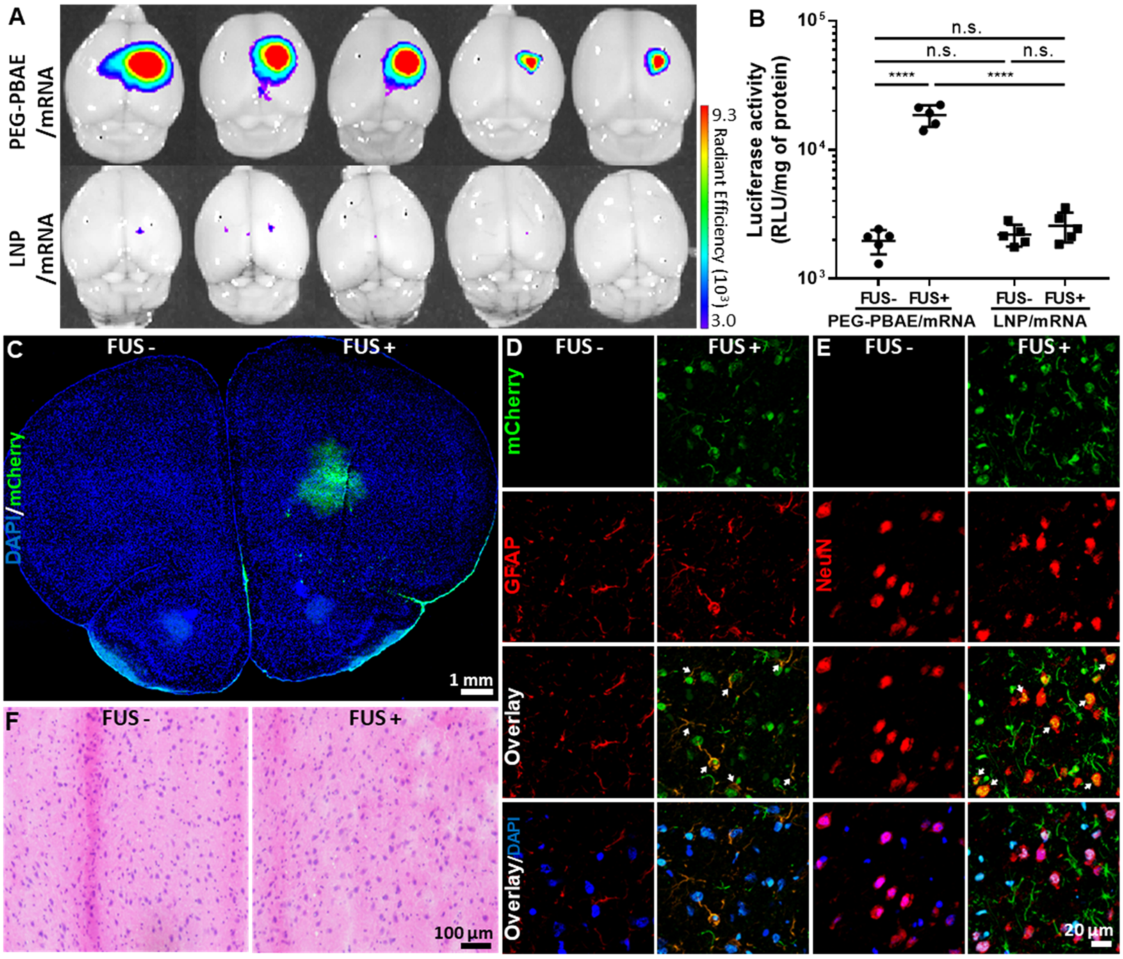


Figure 5. PEG−PBAE NPs, but not clinically used LNPs, carrying mRNA mediate reporter protein production in astrocytes and neurons precisely in the FUS-treated brain region following systemic administration. (A) Bioluminescence images of brains harvested at 24 h postadministration of NPs from animals intravenously treated with PEG−PBAE NPs carrying luciferase-expressing mRNA at an mRNA dose of 0.5 mg/kg and subsequently received a FUS treatment on right striata ($N = 5$ animals per group). (B) Quantification of luciferase activity in the homogenized hemispheres from the brains in panel (A). (C) Representative confocal micrograph of brain harvested at 24 h postadministration of NPs from animals intravenously treated with PEG−PBAE NPs carrying mCherry-expressing mRNA at an mRNA dose of 0.5 mg/kg and subsequently received a FUS treatment on right striata ($N = 5$ animals per group). Blue: nucleus; green: mCherry. Representative confocal micrographs showing reporter mCherry production in untreated or FUS-treated areas of (D) GFAP- and (E) NeuN-stained brains from animals identically treated as in panel (C). Blue: nucleus; green: mCherry; red: astrocyte (GFAP) or neuron (NeuN). (F) Representative H&E-stained histological images of untreated or FUS-treated areas of brains harvested from animals identically treated as in panel (C). n.s.: no significance, $^{****}p < 0.0001$ (one-way ANOVA).

Serum Incubation Does Not Compromise the Ability of PEG−PBAE NPs to Mediate Widespread Transgene Expression in Mouse Brains In Vivo. We tested whether PEG−PBAE NPs retained their ability to efficiently distribute and provide widespread transgene expression in rodent brains[6] after incubation in mouse serum. PEG−PBAE NPs carrying ZsGreen1- or luciferase-expressing pDNA were freshly prepared or incubated in the mouse serum for 30 min and intracranially infused into the right striata of mouse brains at a pDNA dose of 0.1 mg/kg via convection-enhanced delivery. Forty-eight hours after the administration, brain tissues were harvested and sectioned, followed by confocal microscopy of the infusion planes. We found that freshly prepared and serum-incubated PEG−PBAE NPs provided comparably widespread and high-level reporter transgene expression in mouse brains, as revealed by quantitative analysis of confocal micrographs (Figures 3C−E and S2) and tissue homogenate-based luciferase assay (Figure 3F).

FUS-Mediated BBB Opening Enables Dose-Dependent Reporter Transgene Expression in the Brain by Systemically Administered PEG−PBAE NPs. We next investigated whether the combination of FUS-mediated BBB-opening and long-circulating PEG−PBAE/pDNA NPs enabled efficient gene transfer to and reporter transgene expression in mouse brains. PEG−PBAE NPs carrying luciferase-expressing pDNA and MBs were coadministered into male C57BL/6 mice via tail vein injection at doses of 1.25−2.5 mg/kg (pDNA) and 0.3 mL/kg, respectively, and FUS was immediately applied to the right striatum using the FUS parameters described above. Forty-eight hours after the administration, brain tissues were harvested and subjected to whole-brain bioluminescence imaging and tissue homogenate-based luciferase assay. The luminescence signals were observed in the FUS-treated regions within the right brain hemispheres (Figure 4A), and the level of luciferase transgene expression was significantly greater with the higher dose (2.5 mg/kg)

https://doi.org/10.1021/acsnano.4c05270
ACS Nano 2024, 18, 24139−24153

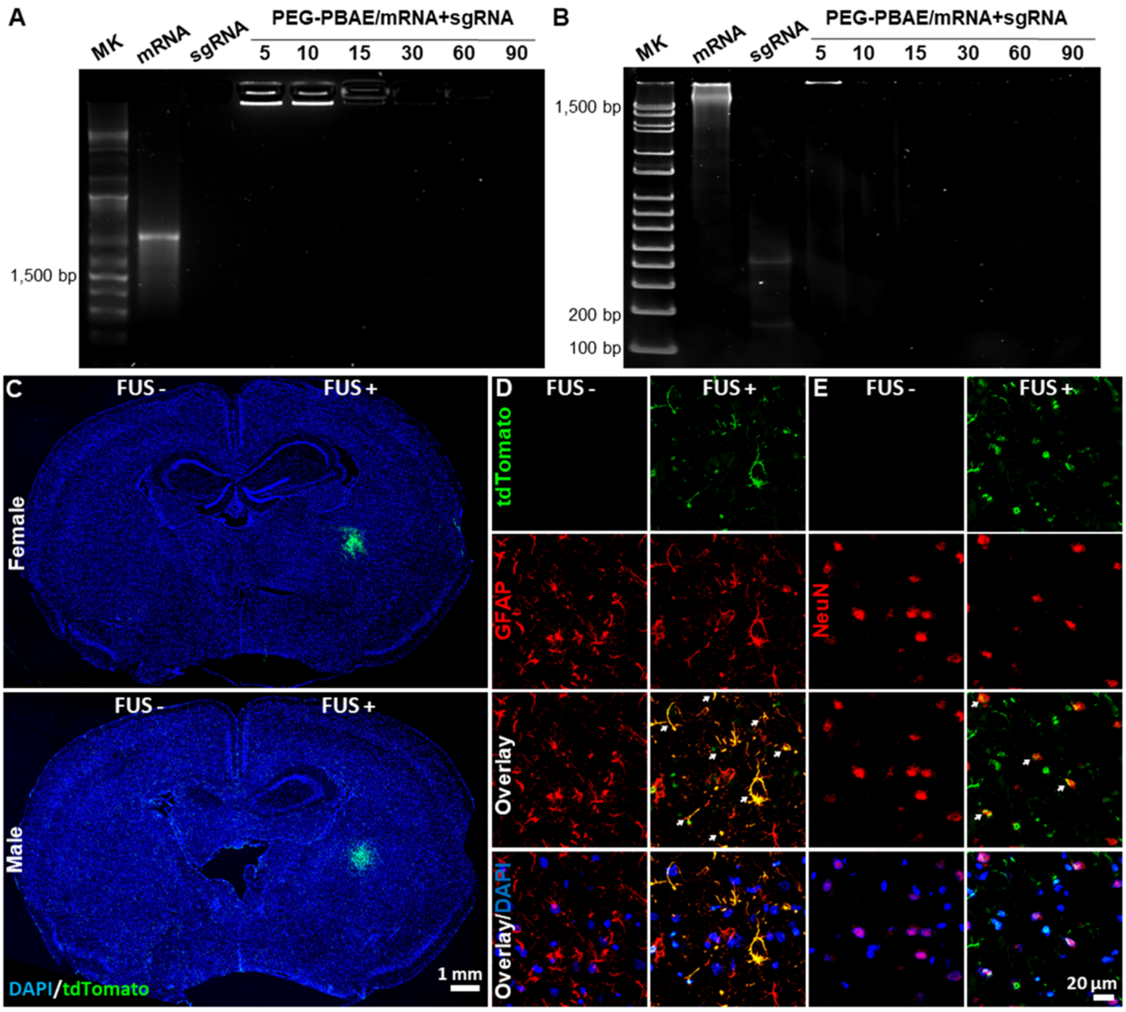


Figure 6. PEG−PBAE NPs carrying Cas9-expressing mRNA and sgRNA targeting the STOP cassette (sgRNA 298) mediate genome editing in astrocytes and neurons precisely in the FUS-treated regions of Ai9 mouse brains following systemic administration. (A) Agarose gel and (B) tris-borate-EDTA (TBE) gel electrophoresis migration assay showing robust copackaging of Cas9-expressing mRNA and sgRNA 298 in PEG−PBAE NPs. Numbers indicate the polymer-to-nucleic-acid weight ratios. MK: molecular weight marker. (C) Representative confocal micrographs of brains harvested at 5 days postadministration of NPs from female (upper) and male (bottom) Ai9 mice intravenously treated with PEG−PBAE NPs carrying Cas9-expressing mRNA and sgRNA 298 at an mRNA and sgRNA dose each of 0.5 mg/kg and subsequently received FUS treatment on right striata (N = 6 animals per group). Blue: nucleus; green: tdTomato. Representative confocal micrographs showing reporter tdTomato production in untreated or FUS-treated areas of (D) GFAP- and (E) NeuN-stained brains from animals identically treated as in panel (C). Blue: nucleus; green: tdTomato; red: astrocyte (GFAP) or neuron (NeuN).

compared to the lower dose (1.25 mg/kg) (Figure 4B). In the subsequent experiment with PEG−PBAE NPs carrying ZsGreen1-expressing pDNA, ZsGreen1 transgene expression was evident in the FUS-treated region (Figure 4C) where the expression was colocalized with GFAP-positive astrocytes and NeuN-positive neurons (Figure 4D,E). As a safe assessment, brain tissues from the treated animals were harvested, sectioned, and stained with hematoxylin and eosin (H&E) for histopathological analysis. A blinded analysis by a board-certified neuropathologist revealed that FUS-treated and contralateral hemispheres were virtually identical with no sign of adverse events at 2 and 14 days postadministration of NPs (Figures 4F and S3).

FUS-Mediated BBB Opening Enables Reporter mRNA Expression by Systemically Administered PEG−PBAE NPs. Excellent colloidal stability (Figure 1E) and stable systemic circulation (Figure 2) observed with PEG−PBAE/mRNA NPs and the successful execution of the systemic pDNA delivery study (Figure 4) collectively provided an optimism that our combined delivery strategy would also work for mRNA. We thus tested FUS-mediated PEG−PBAE/mRNA delivery to the brain using an experimental design identical to the earlier studies with PEG−PBAE/pDNA NPs. PEG−PBAE NPs or LNPs carrying luciferase-expressing mRNA and MBs were coadministrated into male C57BL/6 mice via tail vein injection at doses of 0.5 mg/kg (mRNA) and 0.3 mL/kg, respectively, and FUS was immediately applied to the right striatum. Whole-brain bioluminescence imaging and homogenate-based luciferase assay revealed robust luciferase mRNA expression by PEG−PBAE NPs in the FUS-treated regions at 24 h postadministration of NPs, but the mRNA expression by LNPs was negligible (Figure 5A,B). We also confirmed that PEG−PBAE NPs mediated comparably high mRNA expression in the FUS-treated brain regions at 6 h postadministration, whereas PBAE NPs and LNPs were unable to produce the reporter protein (Figure S4). In contrast, PEG−PBAE NPs exhibited significantly lower hepatic mRNA expression compared to PBAE NPs and LNPs (Figure S4).

https://doi.org/10.1021/acsnano.4c05270
ACS Nano 2024, 18, 24139−24153

The mRNA delivery study was then repeated with PEG−PBAE NPs carrying mCherry-expressing mRNA. We found that mCherry mRNA expression took place in the FUS-treated region (Figure 5C), and the reporter proteins were produced by both GFAP-positive astrocytes and NeuN-positive neurons (Figure 5D). Similar to the observation in the pDNA delivery study above, the combination of FUS-mediated BBB-opening and systemically administered PEG−PBAE/mRNA NPs did not manifest any neuropathological features in the treated brains 1 day after NP administration (Figure 5F).

Combined Delivery Strategy of FUS-Mediated BBB-Opening and Long-Circulating PEG−PBAE NPs Provides a Means to Achieve Somatic Genome Editing in Mouse Brains In Vivo. We next investigated whether our combined delivery strategy could bring about nucleic-acid-based genome editing in the brain. PEG−PBAE NPs were engineered to simultaneously carry Cas9-expressing mRNA and sgRNA 298, and effective packaging of both components was confirmed by gel electrophoresis migration assay (Figure 6A,B). Of note, sgRNA 298 targets the loxP-flanked STOP cassette that prevents the transcription of the tdTomato reporter gene in the Ai9 mouse chromosome.[29] Physicochemical properties (i.e., hydrodynamic diameters and ζ-potentials) of PEG−PBAE NPs carrying both mRNA and sgRNA were confirmed to be comparable to those of PEG−PBAE/pDNA and PEG−PBAE/mRNA NPs (Table 1). PEG−PBAE NP and MBs were then coadministrated into female and male Ai9 mice via tail vein injection at doses of 0.5 mg/kg (mRNA and sgRNA each) and 0.3 mL/kg, respectively, and FUS was immediately applied to the right striatum using the FUS parameters described above. Both female and male Ai9 mice exhibited comparably clear tdTomato expression patterns in the FUS-treated regions within the right striata at 5 days postadministration of NPs (Figure 6C), indicating that the combined delivery strategy successfully disrupted the STOP cassette to mediate somatic genome editing. Similar to the earlier reporter pDNA and mRNA delivery studies, tdTomato expression was evident in both GFAP-positive astrocytes and NeuN-positive neurons (Figure 6D,E).

DISCUSSION

We demonstrate here that long-circulating PEG−PBAE NPs, in conjunction with FUS-mediated BBB opening, promote efficient and safe delivery of various types of nucleic acid payloads precisely to the FUS-treated region within the brain, following systemic administration. Accordingly, the combined delivery strategy provides robust protein production and somatic genome editing in primary brain parenchymal cells in the FUS-treated region in vivo. Importantly, PEG−PBAE NPs exhibit markedly greater accumulation in the FUS-treated brain region than in the liver, while the liver is the primary destination of systemically administered NPs,[30] as shown with un-PEGylated PBAE NPs and clinically used LNPs. Likewise, FUS-mediated BBB opening has been shown to enhance the accumulation of and genome editing by systemically administered adeno-associated virus (AAV) serotype 9 (AAV9) in the brain.[31] AAV9 is a serotype that, unlike other serotypes, possesses ability to traverse the BBB and has been approved by the FDA for the treatment of spinal muscular dystrophy (i.e., Zolgensma).[32] Nevertheless, the amount of AAV9 accumulated in the liver remained 2 orders of magnitude greater than the amount found in the FUS-treated brain region despite the addition of FUS to enhance the brain accumulation.[31] Of note, hepatoxicity is a major and significant concern that limits the widespread use of AAV vectors for the implementation of gene therapy via the systemic route.[33−35] To this end, the combined delivery strategy of FUS-mediated BBB-opening and long-circulating nonviral NPs constitutes an attractive means to enable safe, efficient, and site-specific therapeutic nucleic acid delivery to and therapy in the brain.

Our experimental data demonstrate that the ability of systemically administered nucleic acid delivery NPs to stably circulate and retain colloidal stability in the bloodstream is essential to exploit FUS-mediated BBB opening to partition into the brain parenchyma. We found that PEG−PBAE NPs stably circulated in the bloodstream following systemic administration at least up to 4 h, a previously reported time window of FUS-mediated BBB opening.[36] Further, comparably modest levels of systemically administered PEG−PBAE NPs were found in liver and other major organs, unlike PBAE NPs and LNPs that exhibited conspicuous and preferential liver accumulation. The findings collectively suggest that a significant amount of systemically administered PEG−PBAE NPs was continuously passing through the cerebral capillaries and thus was available for FUS-mediated BBB penetration during the BBB-opening time window. We note that fluorescence signal beyond the background level was observed around the brains of animals treated with poorly circulating PBAE NPs or LNPs carrying fluorescently labeled nucleic acids. However, FUS failed to promote brain penetration and subsequent reporter protein production by these formulations. The finding was likely attributed to the inability of these NPs to retain the particle diameters below the size cutoff of the vascular opening created by FUS to traverse the disrupted BBB. In support of this hypothesis, diffusion rates of ∼70 nm PBAE NPs were over an order of magnitude lower in whole mouse serum than in PBS, which corresponds to a particle diameter increment near to or possibly beyond a micron in serum, as estimated by the Stokes−Einstein relationship.[28] On the other hand, 40- and 100 nm PEG−PS NPs were equally and efficiently accumulated in the FUS-treated brain region following systemic administration, but 200 nm PEG−PS NPs were unable to do so. The data suggest that the size cutoff of the vascular opening established by the FUS parameters employed in this study lies somewhere between 100 and 200 nm, which is far smaller than the size of the serum-incubated PBAE NPs.

Various sonication conditions, including FUS parameters and MB type/concentration, have been explored to achieve safe BBB opening in clinical and preclinical settings.[18,19,37,38] Mechanical index (MI) indicates the power of a single pulse of FUS application, which is defined as the peak negative pressure of the ultrasound wave divided by the square root of the center frequency of a FUS transducer.[39] We here set the MI value to 0.836, which falls in the range of 0.25−1.2 previously employed for FUS-mediated BBB-opening studies.[40,41] Of note, greater MI values generally give rise to greater BBB-opening efficacy but are associated with greater chances of vascular or tissue damage.[42,43] Our histological analysis revealed that a single treatment with SonoVue and 0.836-MI FUS did not cause hemorrhage or neuropathological features in the brain 1, 2, and 14 day(s) after the FUS application. Likewise, repeated treatments with SonoVue and 0.8-MI FUS were previously shown not to induce intracerebral hemorrhage and behavioral change, as determined by histological analysis and modified Irwin's test, respectively.[44] In another study,

https://doi.org/10.1021/acsnano.4c05270
ACS Nano 2024, 18, 24139−24153

however, mild yet noticeable intracranial hemorrhage was observed when animals received a single treatment of SonoVue and 0.85-MI FUS.[45] We note that the comparison should be made with caution as FUS-mediated vascular and/or tissue perturbation can also be affected by other FUS parameters not factored in MI, such as burst duration, pulse repetition frequency, and overall exposure time.[42] Nevertheless, it appears that the MI value utilized in this study may reside around a safely operable borderline, and thus, a more comprehensive safety assessment is warranted in future studies. For example, microglial activation is often observed in the brain upon the implementation of FUS-mediated BBB opening.[46] Although it is generally self-resolved in healthy brains over time,[47,48] such a perturbation brings about a cascade of immune responses[48,49] and has numerous and case-sensitive implications for neurological disorders.[50]

We and others have previously shown that FUS-mediated BBB opening enables and/or enhances delivery of lipid- or polymer-based NPs carrying pDNA to the brain following systemic administration.[51−56] We demonstrate here that systemically administered PEG−PBAE NPs carrying reporter or genome editing (i.e., Cas9) mRNA are efficiently delivered to the brain with the aid of FUS to produce the respective encoded proteins both in astrocytes and neurons, which has been rarely reported, if any. Ogawa et al. recently showed that FUS-mediated BBB opening enhanced the delivery of a specific preclinical LNP formulation carrying reporter mRNA to the brain following systemic administration.[57] However, production of reporter fluorescence proteins was primarily confined to CD31-positive endothelial cells, lacking fluorescence signal overlapping with astrocytes or neurons.[57] The discrepancy observed in these two studies may be at least partially attributed to the ability of PEG−PBAE NPs to efficiently penetrate the brain ECM[6,10] after traversing the BBB transiently opened by FUS to reach brain parenchymal cells. Importantly, we confirmed in this study that colloidal stability and brain-penetrating properties of PEG−PBAE NPs were not compromised after incubation in mouse serum, underscoring the suitability of the formulation for delivery of mRNA and other therapeutic nucleic acids to the brain via the systemic route.

As mentioned above, PEG−PBAE NPs exhibited the capability of stably packaging various types of nucleic acid payloads and delivering them precisely to the FUS-treated brain region. Our microscopic analysis revealed that PEG−PBAE/pDNA and PEG−PBAE/mRNA NPs provided comparable striatal coverage areas of reporter fluorescence protein production by the respective nucleic acid payloads. The outcome was somewhat expected a priori as PEG−PBAE NPs, regardless of the type of payloads, displayed virtually identical physicochemical properties and physiological colloidal stability, the properties that play a pivotal role on particle distribution in the brain.[5,8] However, despite the 5-time lower dose of mRNA compared to pDNA given to animals, the former appeared to provide a greater overall level of reporter protein production, as determined by the tissue homogenate-based luciferase assay. The finding likely reflects that while mRNA is readily translated in the cytosol,[58] pDNA must breach an additional challenging intracellular barrier, nuclear envelope, to produce final protein products.[59] We also found that the striatal coverage area of genome editing in Ai9 mice was smaller than that of the reporter protein production in C57BL/6 mice regardless of the use of identical Cas9- and mCherry-expressing mRNA doses (i.e., 0.5 mg/kg), respectively. The difference is presumably attributed to the necessity of an orchestrated processing of the Cas9 product and sgRNA, beyond mRNA expression, to disrupt the STOP cassette in the Ai9 mouse chromosome for successful genome editing.[60] Nevertheless, notable striatal genome editing was achieved by our combined delivery strategy where the mouse brains were sonicated with a single-element transducer. We expect that a broader coverage of genome editing can be realized, if desired, by using a multielement FUS array that enables BBB opening in a widespread region.[61,62] Indeed, the technique has been utilized for systemic AAV9 delivery to the brain, resulting in more widespread vector distribution and subsequent transgene expression in mouse brains compared to a single-element FUS.[31] We found here that the reporter protein expression patterns in the coronal mouse brain sections were spherical, unlike the ellipsoidal acoustic pressure distribution created by BBB-opening FUS.[63] The discrepancy may reflect that the BBB penetration of PEG−PBAE NPs preferentially took place at the central spherical region of the pressure distribution where the acoustic intensity is greatest and thus the BBB-opening efficacy.[64,65]

CONCLUSIONS

We introduce here a combined delivery strategy that comprehensively addresses a series of biological barriers to achieve safe and efficient delivery of nucleic acids to the brain following systemic administration. Our engineered biodegradable NP platform is capable of packaging various types of nucleic acid payloads and stably circulating upon entry into the bloodstream while resisting liver accumulation and particle aggregation. On the other hand, FUS allows transient BBB opening at a predetermined location within the brain to promote the accumulation of long-circulating NPs as large as 100 nm in diameters in brain parenchyma. Accordingly, we have experimentally validated that this two-pronged delivery strategy mediates site-specific delivery of various nucleic acid payloads to and genome editing in brain parenchymal cells following systemic NP administration. We envision that our approach, upon clinical development, will provide a versatile means to realize systemic nucleic acid therapy for a wide spectrum of brain disorders.

EXPERIMENTAL SECTION

Polymer Synthesis. PBAE polymers were synthesized via Michael addition reactions, as previously described.[5] Briefly, 1,4-butanediol diacrylate (Millipore Sigma, Burlington, MA) and 4-amino-1-butanol (Millipore Sigma) were reacted at a molar ratio of 1.05:1 or 1.1:1 at 90 °C for 20 h. PBAE polymers were then collected by centrifugation/precipitation and washed three times in cold diethyl ether (DE) (Thermo Fisher Scientific, Waltham, MA). After drying under vacuum for 5 days, PBAE polymers were dissolved in dichloromethane (DCM) (Thermo Fisher Scientific) at 100 mg/mL concentration and reacted with 1,4-butanediol diacrylate at a quarter molar equivalent of the amount used for the initial PBAE polymerization reaction to ensure that the two terminal ends of the PBAE polymer chains are capped with acrylate groups. The acrylated PBAE polymers were then purified with cold DE and dried under vacuum for 5 days. The polymers were dissolved in deuterated methanol (Thermo Fisher Scientific) at a 25 mg/mL concentration and analyzed by nuclear magnetic resonance (NMR). Based on the NMR spectrum, the linear structure of acrylated PBAE polymers was confirmed, and the molecular weight (MW) of the polymer synthesized at a molar ratio of 1.05:1 or 1.1:1 was estimated to be 6.0 ± 0.3 or 3.9 ± 0.2 kDa, respectively (Figure S5A,B). In parallel,

https://doi.org/10.1021/acsnano.4c05270
ACS Nano 2024, 18, 24139−24153

the number average MW (polydispersity index) of PBAE polymers synthesized at a molar ratio of 1.05:1 or 1.1:1 was determined by gel permeation chromatography to be 6.7 ± 0.5 kDa (1.37) or 4.3 ± 0.3 kDa (1.31), respectively. To cap the terminal acrylate ends with functional groups, the high-MW (~6 kDa) and low-MW (~4 kDa) acrylated PBAE polymers were dissolved individually in DCM at 100 mg/mL concentration and reacted with 30 mol equiv of 1,11-diamino-3,6,9-trioxaundecane (Millipore Sigma) and 1,2-diaminoethane (Millipore Sigma), respectively, at room temperature for 16 h. The end-capped PBAE polymers were purified with cold DE and dried under vacuum for 5 days. Successful end-capping was confirmed with the complete disappearance of the proton peaks of the acrylate functional group in the NMR spectrum (Figure S5C,D).

To prepare PEG−PBAE polymers, 4 kDa PBAE polymers end-capped with 1,2-diaminoethane were dissolved in DCM at 100 mg/mL concentration and reacted with 3 mol equiv of 5 kDa methoxy PEG-epoxide (Creative PEGWorks, Durham, NC) for 16 h at room temperature. The polymer products were then transferred into a tube of regenerated cellulose membrane having a 10 kDa MW cutoff (MWCO) and extensively dialyzed against methanol (Thermo Fisher Scientific) at 4 °C for 3 days. The polymers were then collected by centrifugation/precipitation, washed three times in cold DE, and dried under vacuum for 5 days. The PEG−PBAE polymer was also analyzed by NMR to confirm the retention of the PBAE backbone structure and PEG conjugation (Figure S5E). PBAE and PEG−PBAE polymers were stored at −80 °C and dissolved in anhydrous dimethyl sulfoxide at 100 mg/mL to be used to formulate NPs carrying various nucleic acid payloads.

Acquisition and Preparation of Nucleic Acids. Luciferase-expressing pDNA driven by a cytomegalovirus (CMV) promoter, pd1GL3-RL, was a kind gift from Professor Alexander M. Klibanov (M.I.T). ZsGreen1-expressing pDNA driven by a CMV promoter was purchased from Clonetech Laboratories (Mountainview, CA). Competent*Escherichia coli*DH5α bacterial cells (New England Biolabs, Ipswich, MA) were transformed with each pDNA. The transformed bacterial cells were placed on an agarose plate containing 50 μg/mL kanamycin (Millipore Sigma) and incubated at 37 °C in a humid chamber. After overnight incubation, a single colony was picked and expanded in 2 mL of sterilized Lennox Broth (LB) (Millipore Sigma) medium containing 50 μg/mL kanamycin. After overnight incubation, 1 mL of the expanded bacterial cells was transferred into 2 L of sterilized LB medium containing 50 μg/mL kanamycin and further incubated overnight. When the optical density at 600 nm wavelength (OD600) of the bacterial culture fell in a range of 3−3.5 as measured by a microplate reader (BioTek, Winooski, VT), bacteria cells were lysed and pDNA was purified using the EndoFree Plasmid Giga Kit (Qiagen, Hilden, Germany) as per the manufacturer's protocol. The final pDNA pellet was resuspended in DNase/RNase-free distilled water (Thermo Fisher Scientific).

The mRNA encoding luciferase, mCherry, or streptococcus pyogene Cas9, modified with 5-methoxyuridine, was purchased from TriLink BioTechnologies (San Diego, CA). The sgRNA 298 modified with the 2′ *O*-methyl analogue on the first and last three bases and 3′ phosphorothioate between the first three and last two bases was purchased from Synthego (Redwood City, CA).

The concentrations of pDNA, mRNA, and sgRNA were measured using a NanoDrop ND-1000 spectrophotometer (NanoDrop Technologies, Wilmington, DE).

Preparation of PEG−PS NPs. Red-fluorescent, carboxylated PS NPs (Thermo Fisher Scientific) sized 40, 100, or 200 nm in diameters were densely coated with 5 kDa methoxy PEG-amine (Creative PEGWorks) via carboxyl-amine coupling reactions, as described in a previous work.[66] Briefly, each PS NP suspension was sonicated for 1 h, washed with DNase/RNase-free distilled water using 100 kDa MWCO Amicon Ultra Centrifugal Filters, and resuspended in 1 mL of DNase/RNase-free distilled water at a PS NP concentration of 10 mg/mL. Methoxy PEG-amine at 5 mol equiv to the particle surface carboxyl groups was added to 40, 100, or 200 nm PS NP suspension, and *N*-hydroxysulfosuccinimide (Millipore Sigma) was added to a final concentration of 7 mM. Subsequently, four volumes of 200 mM borate buffer (pH 8.2) were added to the NP suspension, followed by the addition of 1-ethyl-3-(3-dimethylaminopropyl) carbodiimide (EDC, Millipore Sigma) to a final concentration of 2 mM. PS NP suspensions were reacted at 25 °C for 4 h, washed with DNase/RNase-free distilled water, and centrifuged in 100 kDa MWCO Amicon Ultra Centrifugal Filters to collect PEG−PS NPs. PEG−PS NPs were resuspended at a 50 mg/mL PS NP concentration and stored at 4 °C until use.

Preparation of PBAE, PEG−PBAE NPs, or LNPs Carrying pDNA, mRNA, or a Mixture of mRNA and sgRNA. PBAE or PEG−PBAE NPs were prepared by a formulation method that we have previously reported with minor modifications.[5,67] PBAE NPs were prepared with 6 kDa PBAE polymers, while PEG−PBAE NPs were fabricated with a blend of 6 kDa PBAE and PEG−PBAE polymers at a PBAE weight ratio of 3:2. For the preparation of PEG−PBAE NPs carrying both mRNA and sgRNA, a blend of mRNA and sgRNA at a weight ratio of 1:1 was packaged into the NPs. All NPs were formulated by vigorous mixing of the polymer and nucleic acid solutions at a volumetric ratio of 1:5 and various polymer-to-nucleic-acid weight ratios ranging from 5 to 90. The mixed solution was incubated at room temperature for 30 min for NP assembly. The solution was then transferred into 100 kDa MWCO Amicon Ultra Centrifugal Filters (Millipore Sigma) and centrifuged at 1000*g* for 15 min at 4 °C. The concentrated NP solution was diluted 10 times with DNase/RNase-free distilled water and recentrifuged at 1000*g* for 15 min at 4 °C to remove residual polymers. The process was repeated three times, and the final NP solution was concentrated to a nucleic acid concentration of 1 mg/mL for subsequent experiments.

For the preparation of LNPs, a lipid mixture was prepared with ALC-0315 ionizable lipid, 1,2-distearoyl-*sn*-glycero-3-phosphorylcholine, ALC-0159 PEG lipid, and cholesterol with a molar ratio of 46.3/9.4/1.6/42.7 in absolute ethanol (Thermo Fisher Scientific) at a lipid concentration of 6.32 mg/mL. The mRNA solution was prepared with mRNA in 50 mM sodium acetate (pH 5.0) at an mRNA concentration of 0.21 mg/mL. The lipid mixture and mRNA solutions were mixed at a volume ratio of 1:3 using a NanoAssemblr Ignite (Precision NanoSystems, Canada) at a flow rate of 12 mL/min to formulate LNPs. LNPs were washed with PBS (pH 7.4) using 100 kDa MWCO Amicon Ultra Centrifugal Filters, concentrated to a final mRNA concentration of 1 mg/mL, and stored at 4 °C until use.

Gel Electrophoresis Migration Assay. The amount of PEG−PBAE NPs equivalent to 100 ng of nucleic acids was loaded into the wells of 1% agarose or 4−20% gradient NOVEX TBE gel (Thermo Fisher Scientific). Electrophoresis was conducted at 80 V for 30 min in tris-acetate-EDTA or TBE buffer for agarose or TBE gel, respectively. Subsequently, the agarose gel premade with 1X SYBR Safe DNA Gel Stain (Thermo Fisher Scientific) was imaged using a Chemi-Doc imaging system (Bio-RAD, Hercules, CA). TBE gels were further incubated in TBE buffer containing 1X SYBR Safe DNA Gel Stain at room temperature for 10 min, washed with fresh TBE buffer for another 10 min, and imaged using a Chemi-Doc imaging system.

TEM. PEG−PBAE NPs equivalent to 1 μg of nucleic acids were loaded on a carbon type-B copper grid (Ted Pella, Redding, CA) and air-dried for 6 h. After rinsing with deionized water for 1 min, the sample was stained with UranylLess EM Stain (Electron Microscopy Sciences, Hatfield, PA) for 1 min. The grid was then washed once with deionized water, dried overnight, and imaged with a Hitachi H7600 TEM (Hitachi High-Technologies, Tokyo, Japan).

Physicochemical Characterization and Physiological Stability Assessment. PEG−PBAE NPs were diluted to a nucleic acid concentration of 5 μg/mL in 10 mM NaCl at pH 7.0. The diluted NP solution was transferred into a UV cuvette or a Capillary Zeta cell (Malvern Instruments, Malvern, U.K.), and hydrodynamic diameters or ζ-potentials of NPs, respectively, were measured after a 2 min incubation at room temperature using a Zetasizer Nano ZS (Malvern Instruments, Malvern, U.K.).

To assess the colloidal stability of NPs in a physiological condition in the brain, PBAE or PEG−PBAE NPs equivalent to a nucleic acid concentration of 5 μg/mL were incubated in aCSF (Harvard Apparatus, Holliston, MA) at 37 °C for up to 8 h. At each designated time point, a small volume of the incubated NP solution was transferred into a UV cuvette for the measurement of hydrodynamic diameters using a Zetasizer Nano ZS.

MPT Analysis. For microscopic observation of NPs, pDNA or mRNA were labeled with Cy5 using a Label IT Tracker Intracellular Nucleic Acid Localization Kit (MirusBio, Madison, WI) as per manufacturer's protocol and used to formulate PBAE or PEG−PBAE NPs, as described above.

One microliter of PBAE or PEG−PBAE NPs equivalent to 1 μg of nucleic acids was added to and gently mixed with 30 μL of PBS or whole serum obtained from mouse blood. After 30 min of incubation, the motions of NPs in PBS or mouse serum were recorded with fluorescence video microscopy (Axiovert, Carl Zeiss, Stuttgart, Germany) at a frame rate of 15 frames per second (i.e., 67 ms per frame or 15 frames per second). The movies were then analyzed by using custom-written software in MATLAB (MathWorks, Natick, MA) to determine MSD values.

Assessment of Systemic Circulation and Biodistribution. C57BL/6 mice (6−8-week-old, male) were purchased from Jackson Laboratory (Bar Harbor, ME). Animals were handled in accordance with the guidelines and policies of the Johns Hopkins University Institutional Animal Care and Use Committee. The entire chest and abdomen of animals were shaved with a hair trimmer and hair remover cream. Animals were treated with 0.1 mL of PBAE NPs, PEG−PBAE NPs, or LNPs carrying Cy5-labeled pDNA or mRNA at a nucleic acid dose of 0.5 mg/kg via tail vein injection. The fluorescence intensity of three different NP formulations was virtually identical, regardless of the type of nucleic acid payloads, as confirmed by the fluorometrically measured nucleic acid loading efficiency (Figure S6). Animals were then subjected to whole-body live-animal imaging at 5 min and 0.5, 1, 2, and 4 h postadministration of NPs using an In Vivo Imaging System (IVIS, PerkinElmer, Waltham, MA) at an excitation/emission wavelength of 640/680 nm. Immediately after the final imaging (i.e., 4 h postadministration of NPs), blood samples were harvested from individual animals, which were then subjected to intracardiac perfusion to remove the residual blood. Subsequently, major organs, including brain, heart, liver, kidney, lung, and spleen, were harvested and imaged using an IVIS at the excitation/emission wavelength of 640/680 nm. In parallel, the harvested blood samples were centrifuged at 500*g* for 5 min at 4 °C, and the upper serum layer was collected for the measurement of fluorescence intensity at the excitation/emission wavelength of 640/680 nm using a microplate reader. The percentage of nucleic acids that remained circulating in the bloodstream was calculated based on the estimated average blood volume of 6−8-week-old mice weighed 21.9−25 g (1.876 mL; 0.08 mL/g body weight[68]).

Treatment of Animals with FUS and NPs. The heads of 6−8-week-old male C57BL/6 and female/male B6.Cg-Gt(ROSA)-26Sortm9(CAG-tdTomato)Hze/J (Ai9, strain no. 007909, Jackson Laboratory) mice were shaved with a hair trimmer and hair remover cream. Animals were then treated with 0.15 mL of a mixture of PBAE NPs, PEG−PBAE NPs, or LNPs carrying Cy5-labeled or unlabeled nucleic acids and SonoVue MBs (Bracco Diagnostics, Milano, Italy) at doses of 1.25−0.5 mg/kg (nucleic acid) and 0.3 mL/kg (approximately 1.2×10^6 MBs), respectively, via tail vein injection. Of note, labeled and unlabeled nucleic acids were used for the assessment of particle distribution and reporter protein production, respectively. Animals were immediately anesthetized by isoflurane (Baxter International, Deerfield, IL) and placed on an RK50 stereotactic-guided FUS system (FUS Instruments, Toronto, ON, Canada). FUS was then applied to the right striatum with a single-point sonication using a 515-kHz FUS transducer at a 0.6 MPa peak negative pressure with 10 ms burst length for 2 min at a pulse repetition frequency of 1 Hz.

Assessment of Brain Accumulation and Biodistribution. C57BL/6 mice treated with BBB-opening FUS and systemic NPs carrying Cy5-labeled pDNA were euthanized at 4 h postadministration of NPs. The major organs were then harvested and imaged using an IVIS, as described above. Brains were then separated into two hemispheres: FUS-untreated left and FUS-treated right hemispheres. Subsequently, both brain hemispheres and other major organs were weighed, immersed in reporter lysis buffer (Promega, Madison, WI), and homogenized with a bead-based TissueLyser LT (Qiagen) at a 50/s oscillation rate for 30 min at 4 °C. The lysates were then subjected to three freeze-and-thaw cycles and centrifuged at 12,000*g* for 20 min at 4 °C. The supernatants were collected, and fluorescence intensity was measured at the excitation/emission wavelength of 640/680 nm using a microplate reader.

Intracranial NP Administration and Assessment of Transgene Expression. The heads of 6−8-week-old male C57BL/6 mice were shaved with a hair trimmer and hair remover cream. Surgical procedures were performed using standard sterile surgical techniques. Animals were anesthetized by intraperitoneal injection of a 100-μL mixture of ketamine (100 mg/kg) and xylazine (10 mg/kg). A midline scalp incision was made to expose the coronal and sagittal sutures (the midline), and a burr hole was drilled 2 mm lateral to the sagittal suture and 0.5 mm posterior to the bregma. A Neuros Syringe (Hamilton, Reno, NV) connected to a 33-gauge needle was filled with 2 μL of freshly prepared or serum-incubated PEG−PBAE NPs carrying ZsGreen1- or luciferase-expressing pDNA at a pDNA concentration of 1 mg/mL. The needle was then lowered into a depth of 3 mm from the cranium to target the striatum, and the NP solution was infused at a rate of 0.2 μL/min for 10 min as controlled by a Chemyx Nanojet Injector Module (Chemyx, Stafford, TX). When the infusion was completed, the needle was removed slowly at 1 mm/min to prevent backflow. Following the removal, the skin incision was closed using an Autoclip Wound Closing System (Thermo Fisher Scientific), and a thin layer of bacitracin (Millipore Sigma) was placed over the closed incision.

To determine the distribution of reporter transgene expression, brains of animals intracranially treated with NPs carrying ZsGreen1-expressing pDNA were harvested at 48 h postadministration of NPs, frozen in optimal cutting temperature compound (Thermo Fisher Scientific), and sectioned at a 10 μm thickness using a Leica Cryostat (Leica, Wetzlar, Germany). The brain slices were washed with PBS three times for 1 min each and counterstained with DAPI (Thermo Fisher Scientific) for 5 min. After an additional three PBS washes, the slices were mounted with Fluoromount-G Mounting Medium (Thermo Fisher Scientific) and observed with a confocal laser microscope (Zeiss LSM 710, Carl Zeiss, Stuttgart, Germany).

To determine the overall level of reporter transgene expression, brains of separate animals intracranially treated with NPs carrying luciferase-expressing pDNA were harvested at 48 h postadministration of NPs. The brains were immersed in a reporter lysis buffer (Promega, Madison, WI) and homogenized with a bead-based TissueLyser LT at a 50/s oscillation rate for 30 min at 4 °C. The brain lysates were then subjected to three freeze-and-thaw cycles and centrifuged at 12,000*g* for 20 min at 4 °C. The supernatants were subjected to homogenate-based luciferase assay to measure the luciferase activity in the relative light unit (RLU) using Luciferase Assay System (Promega) and a 20/20n luminometer (Turner Biosystems, Sunnyvale, CA). In parallel, the protein concentrations of the supernatants were quantified by a BCA Protein Assay Kit (Thermo Fisher Scientific). The RLU values were normalized by the protein weight.

Bioluminescence/Fluorescence Imaging and Homogenate-Based Luciferase Assay of Brain Tissues: Assessment of Reporter Protein Production. C57BL/6 mice treated with BBB-opening FUS and systemic NPs carrying luciferase-expressing pDNA or mRNA were euthanized at 48 and 24 h postadministration of NPs, respectively. Brains were then harvested and immersed in 15 mg/mL luciferin substrate solution (Gold Biotechnology, St. Louis, MO) for 1 min, followed by bioluminescence imaging using an IVIS for 1 min. To determine the overall level of luciferase expression, brains were harvested at different time points after NP administration (i.e., 6, 24, or 48 h), separated into two hemispheres, lysed, and subjected to homogenate-based luciferase assay, as described above. For comparison at 6 h postadministration of NPs, other major organs were harvested for luciferase assay.

For microscopic observation of reporter protein production, C57BL/6 mice treated with BBB-opening FUS and systemic NPs

https://doi.org/10.1021/acsnano.4c05270
ACS Nano 2024, 18, 24139−24153

carrying luciferase-expressing pDNA or mRNA were euthanized at 48 or 24 h postadministration of NPs, respectively. Likewise, Ai9 mice treated with BBB-opening FUS and systemic NPs carrying Cas9-expressing mRNA and sgRNA 298 were euthanized at 5 day postadministration of NPs. Brains of euthanized animals were then sectioned, and fluorescence emitted from the reporter proteins, either of ZsGreen1, mCherry, or tdTomato, was visualized with confocal microscopy, as described above. For immunofluorescence staining, the brain slices were washed three times with PBS for 1 min each and fixed with 4% paraformaldehyde (Thermo Fisher Scientific) for 5 min at room temperature. After three additional PBS washes, the slices were immersed in PBS containing 0.1% Triton X-100 (Thermo Fisher Scientific) for 5 min at room temperature. After three additional PBS washes, the slices were incubated in PBS containing 1% bovine serum albumin (BSA) (Thermo Fisher Scientific) for blocking at room temperature for 1 h. After three additional PBS washes, the slices were incubated with anti-GFAP antibody (Cat. #: NB300-141, Novus Biologicals, Centennial, CO) or anti-NeuN antibody (Cat. #: ab177487, Abcam, Cambridge, U.K.) at a 1:200 ratio in PBS containing 1% BSA at 4 °C overnight. After three additional PBS washes, the slices were incubated with donkey antirabbit IgG H&L conjugated with Alexa Fluor 647 at a 1:100 ratio in PBS containing 1% BSA at room temperature for 1 h. After three additional PBS washes, the slices were counterstained with DAPI, mounted with Fluoromount-G Mounting Medium, and observed with a confocal laser microscope.

In Vivo Safety Assessment. Brains harvested from animals treated with BBB-opening FUS and systemic NPs were paraffin-sectioned, and the brain slices were stained with H&E. Histopathological scoring was conducted by a board-certified neuropathologist (C.G.E., M.D., Ph.D.) in a blinded manner for the assessment of local safety profiles of our combined nucleic acid delivery strategy.

Statistical Analysis. Statistical analysis between two groups was conducted using a two-tailed Student's *t* test assuming unequal variances with Welch's correction. If multiple comparisons were involved, one-way ANOVA was conducted followed by a Tukey's multiple comparison test. GraphPad Software (GraphPad Software, La Jolla, CA) was used for these statistical analyses.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsnano.4c05270.

Effect of NP diameter on the ability of systemically administered NPs to traverse the BBB opened by FUS, striatal transgene expression by freshly prepared or serum-incubated PEG−PBAE NPs administered via intracranial CED, neuropathological assessment of brains from animals received FUS treatment and systemic PEG−PBAE NPs at 14 day postadministration of NPs, luciferase activity of the brains and major organs from animals received FUS treatment and PBAE NPs, PEG−PBAE NPs, or LNPs carrying luciferase mRNA at 6 h postadministration of NPs, NMR spectra of PBAE and PEG−PBAE polymers, nucleic acid loading efficacy of PBAE NPs, PEG−PBAE NPs, and LNPs, and physicochemical properties of PEG−PS NPs (PDF)

AUTHOR INFORMATION

Corresponding Author

Jung Soo Suk − *Department of Neurosurgery, School of Medicine, University of Maryland, Baltimore, Maryland 21201, United States; Medicine Institute for Neuroscience Discovery (UM-MIND), School of Medicine, University of Maryland, Baltimore, Maryland 21201, United States; Department of Neurosurgery, School of Medicine, Johns Hopkins University, Baltimore, Maryland 21205, United States; Department of Chemical and Biomolecular Engineering, School of Engineering, Johns Hopkins University, Baltimore, Maryland 21218, United States;* orcid.org/0000-0002-2437-0001; Email: Jsuk@som.umaryland.edu

Authors

Gijung Kwak − *Department of Neurosurgery, School of Medicine, University of Maryland, Baltimore, Maryland 21201, United States; Medicine Institute for Neuroscience Discovery (UM-MIND), School of Medicine, University of Maryland, Baltimore, Maryland 21201, United States;* orcid.org/0000-0001-6908-9638

Angad Grewal − *Department of Neurosurgery, School of Medicine, Johns Hopkins University, Baltimore, Maryland 21205, United States*

Hasan Slika − *Department of Neurosurgery, School of Medicine, Johns Hopkins University, Baltimore, Maryland 21205, United States*

Griffin Mess − *Department of Neurosurgery, School of Medicine, Johns Hopkins University, Baltimore, Maryland 21205, United States*

Haolin Li − *Department of Neurosurgery, School of Medicine, University of Maryland, Baltimore, Maryland 21201, United States; Department of Chemical and Biomolecular Engineering, School of Engineering, Johns Hopkins University, Baltimore, Maryland 21218, United States;* orcid.org/0009-0003-1613-5110

Mohit Kwatra − *Institute for Cell Engineering, School of Medicine and Department of Neurology, School of Medicine, Johns Hopkins University, Baltimore, Maryland 21205, United States*

Alexandros Poulopoulos − *Department of Pharmacology, School of Medicine, University of Maryland, Baltimore, Maryland 21201, United States*

Graeme F. Woodworth − *Department of Neurosurgery, School of Medicine, University of Maryland, Baltimore, Maryland 21201, United States; Medicine Institute for Neuroscience Discovery (UM-MIND), School of Medicine, University of Maryland, Baltimore, Maryland 21201, United States*

Charles G. Eberhart − *Department of Pathology, School of Medicine, Johns Hopkins University, Baltimore, Maryland 21287, United States; Department of Ophthalmology, School of Medicine, Johns Hopkins University, Baltimore, Maryland 21231, United States*

Han Seok Ko − *Institute for Cell Engineering, School of Medicine and Department of Neurology, School of Medicine, Johns Hopkins University, Baltimore, Maryland 21205, United States;* orcid.org/0000-0002-0864-5239

Amir Manbachi − *Department of Neurosurgery, School of Medicine, Johns Hopkins University, Baltimore, Maryland 21205, United States; Department of Biomedical Engineering, School of Medicine, Johns Hopkins University, Baltimore, Maryland 21205, United States*

Justin Caplan − *Department of Neurosurgery, School of Medicine, Johns Hopkins University, Baltimore, Maryland 21205, United States*

Richard J. Price − *Department of Biomedical Engineering, School of Engineering and Applied Science, University of Virginia, Charlottesville, Virginia 22904, United States*

https://doi.org/10.1021/acsnano.4c05270
ACS Nano 2024, 18, 24139−24153

Betty Tyler – *Department of Neurosurgery, School of Medicine, Johns Hopkins University, Baltimore, Maryland 21205, United States*

Complete contact information is available at:
https://pubs.acs.org/10.1021/acsnano.4c05270

Notes

The authors declare the following competing financial interest(s): As an author, Betty Tyler is a co-owner of Accelerating Combination Therapies, LLC and a shareholder of Peabody Pharmaceuticals. One of her patents is licensed to Ashvattha Therapeutics Inc. Other than that, the authors declare that they have no competing interests.

ACKNOWLEDGMENTS

The work reported here was supported by the National Institutes of Health (R01NS111102 and R01NS119609) and the Khatib Brain Tumor Center at the Johns Hopkins University.

REFERENCES

(1) Zhang, Z.; Conniot, J.; Amorim, J.; Jin, Y.; Prasad, R.; Yan, X.; Fan, K.; Conde, J. Nucleic acid-based therapy for brain cancer: Challenges and strategies. *J. Controlled Release* **2022**, *350*, 80–92.

(2) Giotta Lucifero, A.; Luzzi, S.; Brambilla, I.; Guarracino, C.; Mosconi, M.; Foiadelli, T.; Savasta, S. Gene therapies for high-grade gliomas: from the bench to the bedside. *Acta Biomed.* **2020**, *91* (7-S), 32–50.

(3) Zhu, X.; Zhang, Y.; Yang, X.; Hao, C.; Duan, H. Gene Therapy for Neurodegenerative Disease: Clinical Potential and Directions. *Front. Mol. Neurosci.* **2021**, *14*, No. 618171.

(4) Wolak, D. J.; Thorne, R. G. Diffusion of macromolecules in the brain: implications for drug delivery. *Mol. Pharmaceutics* **2013**, *10* (5), 1492–1504.

(5) Negron, K.; Khalasawi, N.; Lu, B.; Ho, C. Y.; Lee, J.; Shenoy, S.; Mao, H. Q.; Wang, T. H.; Hanes, J.; Suk, J. S. Widespread gene transfer to malignant gliomas with In vitro-to-In vivo correlation. *J. Controlled Release* **2019**, *303*, 1–11.

(6) Negron, K.; Zhu, C.; Chen, S. W.; Shahab, S.; Rao, D.; Raabe, E. H.; Eberhart, C. G.; Hanes, J.; Suk, J. S. Non-adhesive and highly stable biodegradable nanoparticles that provide widespread and safe transgene expression in orthotopic brain tumors. *Drug Delivery Transl. Res.* **2020**, *10* (3), 572–581.

(7) Rao, D.; Kwak, G.; Wang, H.; Eberhart, C. G.; Hanes, J.; Suk, J. S. Bioreducible Gene Delivery Platform that Promotes Intracellular Payload Release and Widespread Brain Dispersion. *ACS Biomater. Sci. Eng.* **2023**, *9* (8), 4567–4572.

(8) Negron, K.; Kwak, G.; Wang, H.; Li, H.; Huang, Y. T.; Chen, S. W.; Tyler, B.; Eberhart, C. G.; Hanes, J.; Suk, J. S. A Highly Translatable Dual-arm Local Delivery Strategy To Achieve Widespread Therapeutic Coverage in Healthy and Tumor-bearing Brain Tissues. *Small* **2023**, *19* (11), No. e2207278.

(9) Zhang, C.; Mastorakos, P.; Sobral, M.; Berry, S.; Song, E.; Nance, E.; Eberhart, C. G.; Hanes, J.; Suk, J. S. Strategies to enhance the distribution of nanotherapeutics in the brain. *J. Controlled Release* **2017**, *267*, 232–239.

(10) Mastorakos, P.; Zhang, C.; Song, E.; Kim, Y. E.; Park, H. W.; Berry, S.; Choi, W. K.; Hanes, J.; Suk, J. S. Biodegradable brain-penetrating DNA nanocomplexes and their use to treat malignant brain tumors. *J. Controlled Release* **2017**, *262*, 37–46.

(11) Huang, M.; Gu, X.; Gao, X. 13 - Nanotherapeutic strategies for the treatment of neurodegenerative diseases. In *Brain Targeted Drug Delivery System*; Gao, H.; Gao, X., Eds.; Academic Press, 2019; pp 321–356.

(12) Daneman, R.; Prat, A. The blood-brain barrier. *Cold Spring Harbor Perspect. Biol.* **2015**, *7* (1), No. a020412.

(13) Wu, D.; Chen, Q.; Chen, X.; Han, F.; Chen, Z.; Wang, Y. The blood-brain barrier: structure, regulation, and drug delivery. *Signal Transduction Targeted Ther.* **2023**, *8* (1), No. 217.

(14) Furtado, D.; Bjornmalm, M.; Ayton, S.; Bush, A. I.; Kempe, K.; Caruso, F. Overcoming the Blood-Brain Barrier: The Role of Nanomaterials in Treating Neurological Diseases. *Adv. Mater.* **2018**, *30* (46), No. e1801362.

(15) Liu, H. J.; Xu, P. Strategies to overcome/penetrate the BBB for systemic nanoparticle delivery to the brain/brain tumor. *Adv. Drug Delivery Rev.* **2022**, *191*, No. 114619.

(16) S Hersh, D.; Wadajkar, A. S.; Roberts, N.; Perez, J. G.; Connolly, N. P.; Frenkel, V.; Winkles, J. A.; Woodworth, G. F.; Kim, A. J. Evolving Drug Delivery Strategies to Overcome the Blood Brain Barrier. *Curr. Pharm. Des.* **2016**, *22* (9), 1177–1193.

(17) Jo, S.; Sun, I. C.; Ahn, C. H.; Lee, S.; Kim, K. Recent Trend of Ultrasound-Mediated Nanoparticle Delivery for Brain Imaging and Treatment. *ACS Appl. Mater. Interfaces* **2023**, *15* (1), 120–137.

(18) Gorick, C. M.; Breza, V. R.; Nowak, K. M.; Cheng, V. W. T.; Fisher, D. G.; Debski, A. C.; Hoch, M. R.; Demir, Z. E. F.; Tran, N. M.; Schwartz, M. R.; Sheybani, N. D.; Price, R. J. Applications of focused ultrasound-mediated blood-brain barrier opening. *Adv. Drug Delivery Rev.* **2022**, *191*, No. 114583.

(19) Schoen, S., Jr.; Kilinc, M. S.; Lee, H.; Guo, Y.; Degertekin, F. L.; Woodworth, G. F.; Arvanitis, C. Towards controlled drug delivery in brain tumors with microbubble-enhanced focused ultrasound. *Adv. Drug Delivery Rev.* **2022**, *180*, No. 114043.

(20) Song, K. H.; Harvey, B. K.; Borden, M. A. State-of-the-art of microbubble-assisted blood-brain barrier disruption. *Theranostics* **2018**, *8* (16), 4393–4408.

(21) Fan, C.-H.; Yeh, C.-K. Microbubble-enhanced Focused Ultrasound-induced Blood–brain Barrier Opening for Local and Transient Drug Delivery in Central Nervous System Disease. *J. Med. Ultrasound* **2014**, *22* (4), 183–193.

(22) Burgess, A.; Shah, K.; Hough, O.; Hynynen, K. Focused ultrasound-mediated drug delivery through the blood-brain barrier. *Expert Rev. Neurother.* **2015**, *15* (5), 477–491.

(23) Mehta, R. I.; Carpenter, J. S.; Mehta, R. I.; Haut, M. W.; Wang, P.; Ranjan, M.; Najib, U.; D'Haese, P. F.; Rezai, A. R. Ultrasound-mediated blood-brain barrier opening uncovers an intracerebral perivenous fluid network in persons with Alzheimer's disease. *Fluids Barriers CNS* **2023**, *20* (1), No. 46.

(24) Park, S. H.; Baik, K.; Jeon, S.; Chang, W. S.; Ye, B. S.; Chang, J. W. Extensive frontal focused ultrasound mediated blood-brain barrier opening for the treatment of Alzheimer's disease: a proof-of-concept study. *Transl. Neurodegener.* **2021**, *10* (1), No. 44.

(25) Rezai, A. R.; Ranjan, M.; Haut, M. W.; Carpenter, J.; D'Haese, P. F.; Mehta, R. I.; Najib, U.; Wang, P.; Claassen, D. O.; Chazen, J. L.; Krishna, V.; Deib, G.; Zibly, Z.; Hodder, S. L.; Wilhelmsen, K. C.; Finomore, V.; Konrad, P. E.; Kaplitt, M. Focused ultrasound-mediated blood-brain barrier opening in Alzheimer's disease: long-term safety, imaging, and cognitive outcomes. *J. Neurosurg.* **2023**, *139* (1), 275–283.

(26) Pineda-Pardo, J. A.; Gasca-Salas, C.; Fernandez-Rodriguez, B.; Rodriguez-Rojas, R.; Del Alamo, M.; Obeso, I.; Hernandez-Fernandez, F.; Trompeta, C.; Martinez-Fernandez, R.; Matarazzo, M.; Mata-Marin, D.; Guida, P.; Duque, A.; Albillo, D.; Plaza de Las Heras, I.; Montero, J. I.; Foffani, G.; Toltsis, G.; Rachmilevitch, I.; Blesa, J.; Obeso, J. A. Striatal Blood-Brain Barrier Opening in Parkinson's Disease Dementia: A Pilot Exploratory Study. *Mov. Disord.* **2022**, *37* (10), 2057–2065.

(27) Sonabend, A. M.; Gould, A.; Amidei, C.; Ward, R.; Schmidt, K. A.; Zhang, D. Y.; Gomez, C.; Bebawy, J. F.; Liu, B. P.; Bouchoux, G.; Desseaux, C.; Helenowski, I. B.; Lukas, R. V.; Dixit, K.; Kumthekar, P.; Arrieta, V. A.; Lesniak, M. S.; Carpentier, A.; Zhang, H.; Muzzio, M.; Canney, M.; Stupp, R. Repeated blood-brain barrier opening with an implantable ultrasound device for delivery of albumin-bound paclitaxel in patients with recurrent glioblastoma: a phase 1 trial. *Lancet Oncol.* **2023**, *24* (5), 509–522.

(28) Schuster, B. S.; Ensign, L. M.; Allan, D. B.; Suk, J. S.; Hanes, J. Particle tracking in drug and gene delivery research: State-of-the-art applications and methods. *Adv. Drug Delivery Rev.* **2015**, *91*, 70−91.
(29) Staahl, B. T.; Benekareddy, M.; Coulon-Bainier, C.; Banfal, A. A.; Floor, S. N.; Sabo, J. K.; Urnes, C.; Munares, G. A.; Ghosh, A.; Doudna, J. A. Efficient genome editing in the mouse brain by local delivery of engineered Cas9 ribonucleoprotein complexes. *Nat. Biotechnol.* **2017**, *35* (5), 431−434.
(30) Longmire, M.; Choyke, P. L.; Kobayashi, H. Clearance properties of nano-sized particles and molecules as imaging agents: considerations and caveats. *Nanomedicine* **2008**, *3* (5), 703−717.
(31) Lao, Y. H.; Ji, R.; Zhou, J. K.; Snow, K. J.; Kwon, N.; Saville, E.; He, S.; Chauhan, S.; Chi, C. W.; Datta, M. S.; Zhang, H.; Quek, C. H.; Cai, S. S.; Li, M.; Gaitan, Y.; Bechtel, L.; Wu, S. Y.; Lutz, C. M.; Tomer, R.; Murray, S. A.; Chavez, A.; Konofagou, E. E.; Leong, K. W. Focused ultrasound-mediated brain genome editing. *Proc. Natl. Acad. Sci. U.S.A.* **2023**, *120* (34), No. e2302910120.
(32) Kotulska, K.; Fattal-Valevski, A.; Haberlova, J. Recombinant Adeno-Associated Virus Serotype 9 Gene Therapy in Spinal Muscular Atrophy. *Front. Neurol.* **2021**, *12*, No. 726468.
(33) Hudry, E.; Aihara, F.; Meseck, E.; Mansfield, K.; McElroy, C.; Chand, D.; Tukov, F. F.; Penraat, K. Liver injury in cynomolgus monkeys following intravenous and intrathecal scAAV9 gene therapy delivery. *Mol. Ther.* **2023**, *31* (10), 2999−3014.
(34) High-dose AAV gene therapy deaths *Nat. Biotechnol.* 2020, *388* 910 .
(35) Chand, D.; Mohr, F.; McMillan, H.; Tukov, F. F.; Montgomery, K.; Kleyn, A.; Sun, R.; Tauscher-Wisniewski, S.; Kaufmann, P.; Kullak-Ublick, G. Hepatotoxicity following administration of onasemnogene abeparvovec (AVXS-101) for the treatment of spinal muscular atrophy. *J. Hepatol.* **2021**, *74* (3), 560−566.
(36) Park, J.; Zhang, Y.; Vykhodtseva, N.; Jolesz, F. A.; McDannold, N. J. The kinetics of blood brain barrier permeability and targeted doxorubicin delivery into brain induced by focused ultrasound. *J. Controlled Release* **2012**, *162* (1), 134−142.
(37) Meng, Y.; Pople, C. B.; Lea-Banks, H.; Abrahao, A.; Davidson, B.; Suppiah, S.; Vecchio, L. M.; Samuel, N.; Mahmud, F.; Hynynen, K.; Hamani, C.; Lipsman, N. Safety and efficacy of focused ultrasound induced blood-brain barrier opening, an integrative review of animal and human studies. *J. Controlled Release* **2019**, *309*, 25−36.
(38) Zhang, S.; Zhang, S.; Luo, S.; Tang, P.; Wan, M.; Wu, D.; Gao, W. Ultrasound-assisted brain delivery of nanomedicines for brain tumor therapy: advance and prospect. *J. Nanobiotechnol.* **2022**, *20* (1), No. 287.
(39) Sen, T.; Tufekcioglu, O.; Koza, Y. Mechanical index. *Anatol. J. Cardiol.* **2015**, *15* (4), 334−336.
(40) Chu, P. C.; Chai, W. Y.; Tsai, C. H.; Kang, S. T.; Yeh, C. K.; Liu, H. L. Focused Ultrasound-Induced Blood-Brain Barrier Opening: Association with Mechanical Index and Cavitation Index Analyzed by Dynamic Contrast-Enhanced Magnetic-Resonance Imaging. *Sci. Rep.* **2016**, *6*, No. 33264.
(41) Cammalleri, A.; Croce, P.; Lee, W.; Yoon, K.; Yoo, S. S. Therapeutic Potentials of Localized Blood-Brain Barrier Disruption by Noninvasive Transcranial Focused Ultrasound: A Technical Review. *J. Clin. Neurophysiol.* **2020**, *37* (2), 104−117.
(42) Shin, J.; Kong, C.; Cho, J. S.; Lee, J.; Koh, C. S.; Yoon, M. S.; Na, Y. C.; Chang, W. S.; Chang, J. W. Focused ultrasound-mediated noninvasive blood-brain barrier modulation: preclinical examination of efficacy and safety in various sonication parameters. *Neurosurg. Focus* **2018**, *44* (2), No. E15.
(43) McDannold, N.; Vykhodtseva, N.; Hynynen, K. Blood-brain barrier disruption induced by focused ultrasound and circulating preformed microbubbles appears to be characterized by the mechanical index. *Ultrasound Med. Biol.* **2008**, *34* (5), 834−840.
(44) Tsai, H. C.; Tsai, C. H.; Chen, W. S.; Inserra, C.; Wei, K. C.; Liu, H. L. Safety evaluation of frequent application of microbubble-enhanced focused ultrasound blood-brain-barrier opening. *Sci. Rep.* **2018**, *8* (1), No. 17720.
(45) Wu, S. K.; Chu, P. C.; Chai, W. Y.; Kang, S. T.; Tsai, C. H.; Fan, C. H.; Yeh, C. K.; Liu, H. L. Characterization of Different Microbubbles in Assisting Focused Ultrasound-Induced Blood-Brain Barrier Opening. *Sci. Rep.* **2017**, *7*, No. 46689.
(46) Chen, S.; Nazeri, A.; Baek, H.; Ye, D.; Yang, Y.; Yuan, J.; Rubin, J. B.; Chen, H. A review of bioeffects induced by focused ultrasound combined with microbubbles on the neurovascular unit. *J. Cereb. Blood Flow Metab.* **2022**, *42* (1), 3−26.
(47) Kovacs, Z. I.; Kim, S.; Jikaria, N.; Qureshi, F.; Milo, B.; Lewis, B. K.; Bresler, M.; Burks, S. R.; Frank, J. A. Disrupting the blood-brain barrier by focused ultrasound induces sterile inflammation. *Proc. Natl. Acad. Sci. U.S.A.* **2017**, *114* (1), E75−E84.
(48) Todd, N.; Angolano, C.; Ferran, C.; Devor, A.; Borsook, D.; McDannold, N. Secondary effects on brain physiology caused by focused ultrasound-mediated disruption of the blood-brain barrier. *J. Controlled Release* **2020**, *324*, 450−459.
(49) Kline-Schoder, A. R.; Chintamen, S.; Willner, M. J.; DiBenedetto, M. R.; Noel, R. L.; Batts, A. J.; Kwon, N.; Zacharoulis, S.; Wu, C. C.; Menon, V.; Kernie, S. G.; Konofagou, E. E. Characterization of the responses of brain macrophages to focused ultrasound-mediated blood-brain barrier opening. *Nat. Biomed Eng.* **2024**, *8* (5), 650−663.
(50) Gao, C.; Jiang, J.; Tan, Y.; Chen, S. Microglia in neurodegenerative diseases: mechanism and potential therapeutic targets. *Signal Transduction Targeted Ther.* **2023**, *8* (1), No. 359.
(51) Lin, C. Y.; Hsieh, H. Y.; Chen, C. M.; Wu, S. R.; Tsai, C. H.; Huang, C. Y.; Hua, M. Y.; Wei, K. C.; Yeh, C. K.; Liu, H. L. Non-invasive, neuron-specific gene therapy by focused ultrasound-induced blood-brain barrier opening in Parkinson's disease mouse model. *J. Controlled Release* **2016**, *235*, 72−81.
(52) Mead, B. P.; Mastorakos, P.; Suk, J. S.; Klibanov, A. L.; Hanes, J.; Price, R. J. Targeted gene transfer to the brain via the delivery of brain-penetrating DNA nanoparticles with focused ultrasound. *J. Controlled Release* **2016**, *223*, 109−117.
(53) Lin, C. Y.; Tsai, C. H.; Feng, L. Y.; Chai, W. Y.; Lin, C. J.; Huang, C. Y.; Wei, K. C.; Yeh, C. K.; Chen, C. M.; Liu, H. L. Focused ultrasound-induced blood brain-barrier opening enhanced vascular permeability for GDNF delivery in Huntington's disease mouse model. *Brain Stimul.* **2019**, *12* (5), 1143−1150.
(54) Mead, B. P.; Kim, N.; Miller, G. W.; Hodges, D.; Mastorakos, P.; Klibanov, A. L.; Mandell, J. W.; Hirsh, J.; Suk, J. S.; Hanes, J.; Price, R. J. Novel Focused Ultrasound Gene Therapy Approach Noninvasively Restores Dopaminergic Neuron Function in a Rat Parkinson's Disease Model. *Nano Lett.* **2017**, *17* (6), 3533−3542.
(55) Curley, C. T.; Mead, B. P.; Negron, K.; Kim, N.; Garrison, W. J.; Miller, G. W.; Kingsmore, K. M.; Thim, E. A.; Song, J.; Munson, J. M.; Klibanov, A. L.; Suk, J. S.; Hanes, J.; Price, R. J. Augmentation of brain tumor interstitial flow via focused ultrasound promotes brain-penetrating nanoparticle dispersion and transfection. *Sci. Adv.* **2020**, *6* (18), No. eaay1344.
(56) Yue, P.; Miao, W.; Gao, L.; Zhao, X.; Teng, J. Ultrasound-Triggered Effects of the Microbubbles Coupled to GDNF Plasmid-Loaded PEGylated Liposomes in a Rat Model of Parkinson's Disease. *Front. Neurosci.* **2018**, *12*, 222.
(57) Ogawa, K.; Kato, N.; Yoshida, M.; Hiu, T.; Matsuo, T.; Mizukami, S.; Omata, D.; Suzuki, R.; Maruyama, K.; Mukai, H.; Kawakami, S. Focused ultrasound/microbubbles-assisted BBB opening enhances LNP-mediated mRNA delivery to brain. *J. Controlled Release* **2022**, *348*, 34−41.
(58) Wu, B.; Eliscovich, C.; Yoon, Y. J.; Singer, R. H. Translation dynamics of single mRNAs in live cells and neurons. *Science* **2016**, *352* (6292), 1430−1435.
(59) Bai, H.; Lester, G. M. S.; Petishnok, L. C.; Dean, D. A. Cytoplasmic transport and nuclear import of plasmid DNA. *Biosci. Rep.* **2017**, *37* (6), No. BSR20160616.
(60) Davis, A. J.; Chen, D. J. DNA double strand break repair via non-homologous end-joining. *Transl. Cancer Res.* **2013**, *2* (3), 130−143.

(61) Batts, A. J.; Ji, R.; Noel, R. L.; Kline-Schoder, A. R.; Bae, S.; Kwon, N.; Konofagou, E. E. Using a rapid alternating steering angles pulse sequence to evaluate the impact of theranostic ultrasound-mediated ultra-short pulse length on blood-brain barrier opening volume and closure, cavitation mapping, drug delivery feasibility, and safety. *Theranostics* **2023**, *13* (3), 1180−1197.

(62) Batts, A.; Ji, R.; Kline-Schoder, A.; Noel, R.; Konofagou, E. Transcranial Theranostic Ultrasound for Pre-Planning and Blood-Brain Barrier Opening: A Feasibility Study Using an Imaging Phased Array In Vitro and In Vivo. *IEEE Trans. Biomed. Eng.* **2022**, *69* (4), 1481−1490.

(63) Yoo, S. S.; Bystritsky, A.; Lee, J. H.; Zhang, Y.; Fischer, K.; Min, B. K.; McDannold, N. J.; Pascual-Leone, A.; Jolesz, F. A. Focused ultrasound modulates region-specific brain activity. *NeuroImage* **2011**, *56* (3), 1267−1275.

(64) Seo, H.; Huh, H.; Lee, E. H.; Park, J. Numerical Evaluation of the Effects of Transducer Displacement on Transcranial Focused Ultrasound in the Rat Brain. *Brain Sci.* **2022**, *12* (2), 216.

(65) Choi, M.; Jang, M.; Yoo, S. S.; Noh, G.; Yoon, K. Deep Neural Network for Navigation of a Single-Element Transducer During Transcranial Focused Ultrasound Therapy: Proof of Concept. *IEEE J. Biomed. Health Inf.* **2022**, *26* (11), 5653−5664.

(66) Nance, E. A.; Woodworth, G. F.; Sailor, K. A.; Shih, T. Y.; Xu, Q.; Swaminathan, G.; Xiang, D.; Eberhart, C.; Hanes, J. A dense poly(ethylene glycol) coating improves penetration of large polymeric nanoparticles within brain tissue. *Sci. Transl. Med.* **2012**, *4* (149), No. 149ra119.

(67) Kim, N.; Kwak, G.; Rodriguez, J.; Livraghi-Butrico, A.; Zuo, X.; Simon, V.; Han, E.; Shenoy, S. K.; Pandey, N.; Mazur, M.; Birket, S. E.; Kim, A.; Rowe, S. M.; Boucher, R.; Hanes, J.; Suk, J. S. Inhaled gene therapy of preclinical muco-obstructive lung diseases by nanoparticles capable of breaching the airway mucus barrier. *Thorax* **2022**, *77* (8), 812−820.

(68) Harkness, J. E.; Turner, P. V.; VandeWoude, S.; Wheler, C. L. *Harkness and Wagner's Biology and Medicine of Rabbits and Rodents*, 5th ed.; John Wiley & Sons, 2010.

https://doi.org/10.1021/acsnano.4c05270
ACS Nano 2024, 18, 24139−24153